

END TO END STATISTICAL PROGRAMMING AND TRANSFORMATION OF
RAW DATA TO CDISC STANDARDS (SDTM, ADAM, TLFS) FOR SYNTHETIC
ONCOLOGY CLINICAL TRIAL DATA USING SAS ONDEMAND.

A DISSERTATION SUBMITTED TO



Savitribai Phule Pune University, Pune
IN THE PARTIAL FULFILLMENT OF THE DEGREE OF
MASTER OF PHARMACY
IN
DEPARTMENT OF PHARMACOLOGY

SUBMITTED BY
Mr. AHER SHAILENDRA SHAINATH
(Seat No: 10492)
(FINAL YEAR M.PHARM)

UNDER THE GUIDANCE OF
Dr. AMAN B. UPAGANLAWAR, M.Pharm., Ph.D., M.B.A., FAMS



DEPARTMENT OF PHARMACOLOGY
DR. N.J. PAULBUDHE COLLEGE OF PHARMACY
SHANISIWAR NAGAR, YASANT TEKDI, AHILYANAGAR-414001
2025-2026

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SHANISHWAR NAGAR,VASANT TEKDI AHILYANAGAR-414001

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DECLARATION BY STUDENT

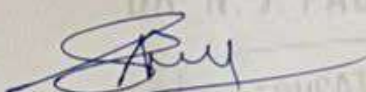
I, Mr. Aher Shailendra Shainath, hereby declare that the dissertation entitled "**End-To-End Statistical Programming And Transformation Of Raw Data To CDISC Standards (SDTM, ADaM, TLF) For Synthetic Oncology Clinical Trial Data Using SAS OnDemand**" submitted to **Dr. N. J. Paulbudhe College of Pharmacy**, affiliated with **Savitribai Phule Pune University, Pune**, in partial fulfillment of the requirements for the award of the **Master of Pharmacy (M.Pharm)** degree under **Project Work (MPL 402)** for the **Academic Year 2025-2026**, is the result of my original research work carried out under the guidance and supervision of **Dr. AMAN B. UPAGANLAWAR**, Department of Pharmacology.

I further declare that this dissertation has not been submitted, either in whole or in part, to any other university or institution for the award of any degree, diploma, fellowship, or any other academic qualification.

I also affirm that all sources of information and literature used in this work have been duly acknowledged and cited wherever appropriate.

Place: Ahilyanagar

Date: 10-08-2026


Signature of the Candidate

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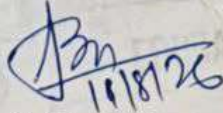
This is to certify that the project entitled **“END-TO-END STATISTICAL PROGRAMMING AND TRANSFORMATION OF RAW DATA TO CDISC STANDARDS (SDTM, ADAM, TLF) FOR SYNTHETIC ONCOLOGY CLINICAL TRIAL DATA USING SAS ONDEMAND”** is a bona fide research work carried out by **Mr. Aher Shailendra Shainath** under my guidance and supervision in the **Department of Pharmacology, Dr. N. J. Paulbudhe College of Pharmacy**, affiliated with **Savitribai Phule Pune University, Pune**.

The work embodied in this dissertation is original and has been completed by the candidate in partial fulfillment of the requirements for the award of the **Master of Pharmacy (M.Pharm)** degree under **Project Work (MPL 402)** during the **Academic Year 2025–2026**. To the best of my knowledge, this work has not been submitted, either in part or in full, to any other university or institution for the award of any degree or diploma.

Place: Ahilyanagar

Date: 11-08-2026

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The dissertation fulfills the requirements prescribed for Project Work (MPL 402) of the Final Year Master of Pharmacy (M.Pharm) programme for the Academic Year 2025-2026.

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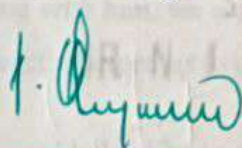
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The dissertation has been completed under the guidance of Dr. AMAN B. UPAGANLAWAR, and is submitted in partial fulfillment of the requirements for the award of the Master of Pharmacy (M.Pharm) degree under Project Work (MPL 402) during the Academic Year 2025-2026.

The work presented is worthy of submission to Savitribai Phule Pune University, Pune for evaluation.

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We sincerely and whole heartedly thank my Research Supervisor, **Dr. AMAN B. UPAGANLAWAR** for his valuable guidance, motivation, and direction, which helped us at all times during this project work. We consider ourselves privileged to have worked under his guidance, as he always shared his vast experience so kindly and patiently in spite of his busy schedule, along with his unfailing advice and constant encouragement during the project work.

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We would also like to thank our colleagues for their help during this work. Lastly but not least, we express our sincere thanks to our lovable parents and family members, without whose support this work was not possible.



Mr. AHER SHAILENDRA SHAINATH

ABSTRACT

The present study focuses on the development and implementation of an end-to-end statistical programming framework for synthetic clinical trial data in oncology, with specific application to advanced non-small cell lung cancer (NSCLC). The study simulates a Phase III randomized, double-blind clinical trial (ONCO-PEMBRO-500), evaluating the safety and efficacy of Pembrolizumab.

DISSERTATION WORK REPORT APPROVAL FOR M.PHARM

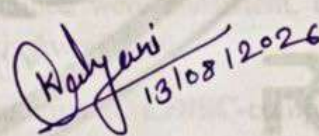
The primary objective of this project is to demonstrate a reproducible and transparent data workflow aligned with global regulatory standards, involving the transformation of raw clinical trial data into CDISC-standardized formats.

This is to certify that the project work entitled “End-to-End Statistical Programming and Transformation of Raw Data to CDISC Standards (SDTM, ADaM, TLF) for Synthetic Oncology Clinical Trial Data Using SAS OnDemand”, carried out by Mr. Aher Shailendra Shainath, has been examined and approved for consideration towards the successful completion of Project Work (MPL 402) of the Final Year Master of Pharmacy (M.Pharm), Semester IV, for the Academic Year 2025–2026, under Savitribai Phule Pune University, Pune.

Statistical analysis was performed using SAS OnDemand for Academics, ensuring industry-standard programming practices. Key outputs included demographic summaries, treatment-emergent adverse event analysis, Kaplan-Meier survival analysis, Overall Survival (OS) and Progression-Free Survival (PFS), and Best Overall Response (BOR) evaluation.

The study adhered to Good Clinical Practice (GCP) and Good Programming Practices (GPP) using Independent External Programming (IEP) where critical analyses and program code were executed independently on the same datasets. The final results were consistent, confirming the robustness of the statistical framework.

EXAMINERS :


1. Dr. Kalyani Autade

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Date: 13-08-2026

Place : Ahilyanagar

ABSTRACT

The present study focuses on the development and implementation of an end-to-end statistical programming framework for synthetic clinical trial data in oncology, with specific application to advanced non-small cell lung cancer (NSCLC). The study simulates a Phase III randomized, double-blind clinical trial (ONCO-PEMBRO-500), evaluating the safety and efficacy of Pembrolizumab-based immunotherapy in comparison with standard chemotherapy.

The primary objective of this project is to demonstrate a complete clinical data workflow aligned with global regulatory standards, involving the transformation of raw clinical datasets into CDISC-compliant formats, namely SDTM (Study Data Tabulation Model) and ADaM (Analysis Data Model), followed by statistical analysis and generation of Tables, Listings, and Figures (TLFs).

Synthetic datasets were systematically designed to replicate real-world clinical trial structures, incorporating key domains such as Demographics (DM), Adverse Events (AE), Laboratory Data (LB), and Tumor Response domains (TU, TR, RS). These datasets were further processed to generate analysis-ready ADaM datasets including ADSL, ADAE, ADLB, ADTTE, and ADRESP.

Statistical analyses were performed using SAS OnDemand® for Academics, ensuring industry-standard programming practices. Key outputs included demographic summaries, treatment-emergent adverse event analysis, Kaplan–Meier survival analysis for Overall Survival (OS) and Progression-Free Survival (PFS), and Best Overall Response (BOR) evaluation.

To ensure accuracy and reliability, Quality Control (QC) was implemented using Independent Double Programming (IDP), where production and validation codes were executed independently on the same datasets. The results obtained were consistent, confirming the robustness and correctness of the statistical programming workflow.

This study successfully demonstrates that a fully automated, CDISC-compliant statistical programming pipeline can be developed using synthetic data, ensuring traceability, reproducibility, and regulatory readiness. The project highlights the critical role of SAS programming in modern clinical research and serves as a strong foundation for real-world clinical trial data analysis and regulatory submissions.

Keywords: Clinical SAS Programming, CDISC, SDTM, ADaM, TLFs, Oncology, Pembrolizumab, Kaplan-Meier Analysis, Quality Control, Synthetic Data.

TABLE OF CONTENTS

SR NO.	PARTICULARS		PAGE NO.
1.	Title Page		1
2.	Certificate		2
3.	Thesis Approval		3
4.	Acknowledgement		4
6.	Declaration		5
7.	Abstract		6
8.	Table of Contents		7-10
9.	List of Abbreviations		11
10.	List of Tables		12
11.	List of Figures		13
CHAPTER 01	INTRODUCTION		15
	1.1	Disease Overview	16
	1.2	Drug Overview: Pembrolizumab	16
	1.3	Clinical Data Challenges	17
	1.4	Need for SAS and CDISC Standards	17
CHAPTER : 02	LITERATURE REVIEW & REGULATORY FRAMEWORK		18
	2.1	Introduction	19
	2.2	KEYNOTE-024 Study Reference	20
	2.3	CDISC Standards in Clinical Research	20
	2.4	Regulatory Authorities and Global Compliance	21
	2.5	“No SAS = No Approval” Concept	21
	2.6	Digital Appendix – A: Protocol Reference	22
CHAPTER : 03	AIM, OBJECTIVES & HYPOTHESIS		23
	3.1	Aim of the Study	24

	3.2	Objectives of the Study	24
	3.3	Research Hypothesis	25
	3.4	Hypothesis Justification	25
CHAPTER : 04	STUDY DESIGN & METHODOLOGY		26
	4.1	STUDY DESIGN	27
	4.2	METHODOLOGY	29
CHAPTER : 05	WHY SYNTHETIC DATA		33
	5.1	Introduction	34
	5.2	Data Privacy and Regulatory Compliance	35
	5.3	Technical Validation of SAS Programming	35
CHAPTER : 06	DATA DESIGN & RAW DATA PREPARATION		36
	6.1	Introduction	37
	6.2	Mapping Sheet Design	38
	6.3	Representative Mapping Tables	38
	6.4	Raw Data Creation (CDM Export Simulation)	40
	6.5	Digital Appendix – B: Domain Mapping Sheet	42
	6.6	Appendix – C : Raw Data (CDM Export Files	43
CHAPTER : 07	SDTM IMPLEMENTATION		44
	7.1	Introduction	45
	7.2	SDTM Overview	45
	7.3	SDTM Domains Implemented	45
	7.4	SDTM Dataset Implementation Using SAS	46
	7.5	SDTM Dataset Verification	47
	7.6	Digital Appendix – D : SDTM Datasets	47
CHAPTER : 08	ADaM ANALYSIS DESIGN & DERIVATION		48
	8.0	Introduction	49

	8.1	ADaM Overview & Derivation	49	
	8.2	ADaM Datasets Implemented	50	
	8.3	ADaM Analysis Design	50	
	8.4	Derivation Specification	50	
	8.5	Digital Appendix – E: ADaM Analysis Design	54	
CHAPTER : 09		ADaM DATASET IMPLEMENTATION	55	
	9.0	Introduction	56	
	9.1	ADaM Datasets Implemented	56	
	9.2	ADaM Dataset Implementation	57	
	9.3	Digital Appendix – F: ADaM Datasets	58	
CHAPTER : 10		TLF SPECIFICATION	59	
	10.0	Introduction	60	
	10.1	TLF Output Blueprint	60	
	10.2	ADaM to TLF Mapping	61	
CHAPTER : 11		STATISTICAL ANALYSIS & TLF GENERATION	62	
	11.0	Introduction	63	
	11.1	Statistical Analysis Overview	63	
	11.2	TLF Generation Using SAS	64	
	11.3	Consolidated Output View	68	
	11.4	Traceability & Data Flow	68	
	11.5	Digital Appendix – G: TLF Statistical Output	69	
CHAPTER : 12		QUALITY CONTROL & VALIDATION	70	
	12.1	Introduction	71	
	12.2	Validation Methodology : Independent Double Programming (IDP)	71	
	12.3	QC Scope and Validation Strategy	72	
	12.4	QC Results & Validation Proof	73	
	12.5	Final QC Summary & Validation Outcome	76	

	12.6	Digital Appendix – H: Quality Control & Validation	77	
	12.7	Summary	77	
CHAPTER : 13	RESULTS & DISCUSSION		78	
	13.1	Introduction	79	
	13.2	Demographic & Baseline Characteristics	79	
	13.3	Treatment Exposure Analysis	80	
	13.4	Safety Analysis (Adverse Events)	81	
	13.5	Efficacy Analysis (Survival Analysis)	82	
	13.6	Response Analysis (BOR)	83	
	13.7	Overall Pipeline Performance	84	
	13.8	Discussion	84	
	13.9	Summary	84	
CHAPTER : 14	CSR READINESS & OUTPUT STRUCTURE		85	
	14.1	CSR Overview	86	
	14.2	CSR Section Mapping	88	
	14.3	SAS Output Structure & CSR Readiness	87	
CHAPTER : 15	TOOLS & TECHNOLOGY ENVIRONMENT		88	
	15.1	Introduction	89	
	15.2	Statistical Programming Environment (SAS)	89	
	15.3	Documentation & Reporting Tools	89	
	15.4	Operating System & Execution Environment	90	
	15.5	Integrated Workflow Environment	90	
CHAPTER : 16	CONCLUSION		91	
	16.1	Introduction	92	
	16.2	End-to-End Workflow Success	92	
	16.3	CDISC Compliance & Data Standardization	92	

	16.4	Statistical Analysis & TLF Generation	93
	16.5	Quality Control & Validation	94
	16.6	Hypothesis Validation & Analytical Accuracy	94
CHAPTER : 17	FUTURE SCOPE & INDUSTRIAL SIGNIFICANCE		95
	17.1	Introduction	96
	17.2	Application in Real Clinical Trials	95
	17.3	Automation & Scalability	95
	17.4	Regulatory Submission Readiness	96
	17.5	Industrial Significance	96
	17.6	Future Enhancements	97
REFERENCE			98
FINAL DIGITAL APPENDIX SUMMARY			99
THANK YOU			100

LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
NSCLC	Non-Small Cell Lung Cancer
CRF	Case Report Form
CSR	Clinical Study Report
QC	Quality Control
IDP	Independent Double Programming
ITT	Intent-To-Treat
SAF	Safety Population
EFF	Efficacy Population
CDISC	Clinical Data Interchange Standards Consortium
SDTM	Study Data Tabulation Model

ADaM	Analysis Data Model
CDASH	Clinical Data Acquisition Standards Harmonization
FDA	Food and Drug Administration
EMA	European Medicines Agency
PMDA	Pharmaceuticals and Medical Devices Agency
CDSCO	Central Drugs Standard Control Organization
SAS	Statistical Analysis System
TLF	Tables, Listings, and Figures
CSV	Comma-Separated Values
XPT	Transport File Format
PROC	Procedure (SAS Programming)
DM	Demographics
AE	Adverse Events
EX	Exposure
LB	Laboratory Data
VS	Vital Signs
TU	Tumor Identification
TR	Tumor Response
RS	Response Status
DS	Disposition
SUPPDM	Supplemental Demographics
SUPPAE	Supplemental Adverse Events
SUPPEX	Supplemental Exposure
SUPPLB	Supplemental Laboratory

LIST OF TABLES

SR. NO.	TABLE TITLE
1	Table 4.1: Protocol Comparison (Reference vs Simulated Study)
2	Table 6.1: Core DM Variables Mapping (Domain: DM – Demographics)
3	Table 6.2: SUPPDM Mapping (Height, Weight, Smoking)
6	Table 8.1: ADSL Dataset Derivation Specification – Dataset Overview
7	Table 8.2: ADSL Dataset – Population Definition
8	Table 10.1: ADaM to TLF Mapping – Tables
9	Table 10.2: ADaM to TLF Mapping – Listings
10	Table 10.3: ADaM to TLF Mapping – Figures
11	Table 11.1: Demographic and Baseline Characteristics (Final Output)
12	Table 11.8: Traceability and Data Flow Summary
13	Table 12.1: Final QC Summary and Validation Outcome
14	Table 13.1: Demographic and Baseline Characteristics (Final Results)
15	Table 13.2: Treatment Exposure Summary (Final Results)
16	Table 13.3: Summary of Treatment-Emergent Adverse Events

LIST OF FIGURES

SR. NO.	FIGURE TITLE
1	Figure 1.1: Lung Cancer (NSCLC) Disease Overview Images
2	Figure 1.2: Mechanism of Action of Pembrolizumab
3	Figure 2.1: Reference Clinical Trial Protocol Snapshot
4	Figure 2.2: ClinicalTrials.gov Study Design Overview
5	Figure 6.1: DM Raw Dataset (CSV Format – Excel View)

6	Figure 6.2: AE Raw Dataset (CSV Format – Excel View)
7	Figure 6.3: SAS Output After Import of Raw Data
8	Figure 7.1: DM Domain – SAS Code, Log, and Output Screenshot
9	Figure 7.2 : SAS Library View Showing All SDTM Datasets
10	Figure 9.1: ADSL Dataset – SAS Code, Log, and Output Screenshot
11	Figure 9.2: SAS Library View Showing All ADaM Datasets
12	Figure 11.1: Kaplan–Meier Plot – Overall Survival (OS)
14	Figure 11.3: Bar Chart – Best Overall Response (BOR)
15	Figure 11.4: Consolidated TLF Output View in SAS Environment
16	Figure 12.1: Quality Control & Validation Strategy (IDP Workflow)
17	Figure 12.2: QC Comparison for Table 1 (Demographics)
18	Figure 12.4: QC Comparison for Listing 1 (Subject Listing)
19	Figure 12.6: QC Comparison for Figure 1 (Overall Survival)
20	Figure 13.1: Kaplan–Meier Plot of Overall Survival (Final Result)
21	Figure 13.2: Kaplan–Meier Plot of Progression-Free Survival (Final
22	Figure 13.3: Bar Chart of Best Overall Response (Final Result)
23	Figure 14.1: SAS Output Structure Showing Organized TLF Files

CHAPTER : 01

INTRODUCTION

1.1 DISEASE OVERVIEW : (NON-SMALL CELL LUNG CANCER - NSCLC)

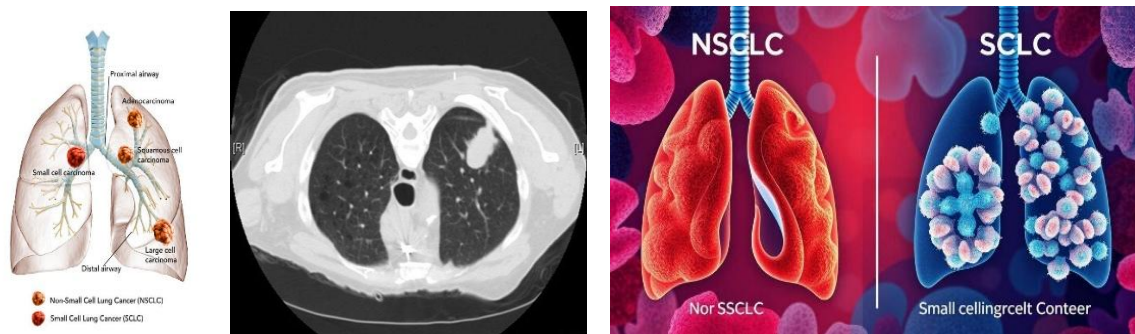


Figure 1.1: Overview of Non-Small Cell Lung Cancer (NSCLC) and Lung Anatomy

Non-Small Cell Lung Cancer (NSCLC) is the most common type of lung cancer, accounting for approximately 85% of all lung cancer cases worldwide. It is a leading cause of cancer-related mortality due to its late-stage diagnosis and aggressive progression.

NSCLC includes several histological subtypes such as adenocarcinoma, squamous cell carcinoma, and large cell carcinoma. Patients are often diagnosed at advanced stages (Stage III or IV), where treatment options become limited and survival rates are significantly reduced.

1.2 DRUG OVERVIEW : PEMBROLIZUMAB

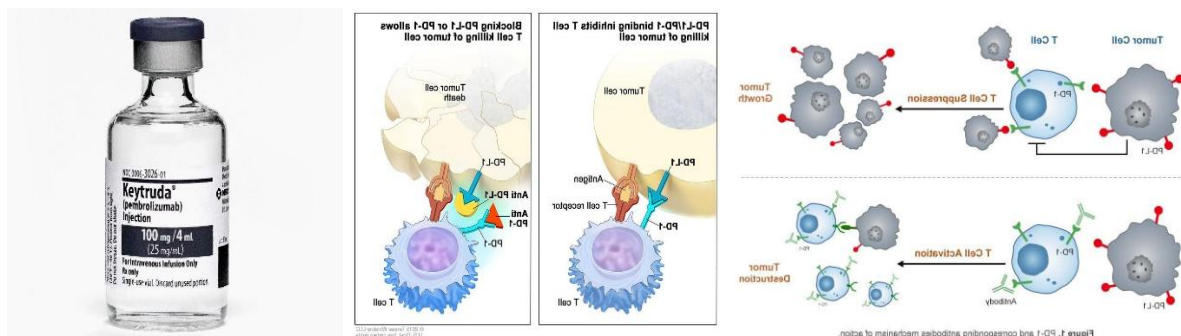


Figure 1.2: Mechanism of Action and Structure of Pembrolizumab Immunotherapy

inhibitor that targets the Programmed Death-1 (PD-1) receptor on T-cells. By blocking the interaction between PD-1 and its ligands (PD-L1/PD-L2), pembrolizumab enhances the immune system's ability to recognize and destroy cancer cells.

It has been widely used in the treatment of advanced NSCLC and has demonstrated significant improvement in overall survival and progression-free survival in multiple clinical trials. Due to its clinical relevance, pembrolizumab is selected as the reference drug model in this study to simulate realistic oncology trial data.

In this project, pembrolizumab is not evaluated for clinical efficacy, but rather used as a framework to generate synthetic clinical trial datasets for statistical programming and CDISC implementation.

1.3 CLINICAL DATA CHALLENGES

Clinical trials generate massive volumes of complex, heterogeneous, and multi-domain data, including patient demographics, laboratory results, adverse events, tumor assessments, and survival outcomes. Managing and analyzing such large datasets manually is inefficient, error-prone, and time-consuming.

Key challenges include:

- Data inconsistency and lack of standardization across sources
- Difficulty in integrating multi-domain datasets
- Increased risk of human error in manual analysis
- Complexity in generating accurate statistical outputs
- Need for traceability and reproducibility of results

These challenges highlight the necessity of automated and standardized data processing systems in clinical research.

1.4 NEED FOR SAS AND CDISC STANDARDS

To address the complexities of clinical data analysis, the pharmaceutical industry widely adopts SAS (Statistical Analysis System) as a standard tool for statistical programming. SAS provides robust capabilities for data manipulation, statistical analysis, and generation of Tables, Listings, and Figures (TLFs).

In parallel, CDISC (Clinical Data Interchange Standards Consortium) standards ensure uniform structure and format of clinical trial data. The two key CDISC models are:

- SDTM (Study Data Tabulation Model): Standardizes raw clinical data for regulatory submission
- ADaM (Analysis Data Model): Provides analysis-ready datasets for statistical evaluation

The integration of SAS with CDISC standards enables efficient, accurate, and reproducible clinical data analysis workflows.

CHAPTER : 02

LITERATURE REVIEW & REGULATORY FRAMEWORK

2.1 INTRODUCTION

Clinical trial data analysis has evolved significantly with the advancement of statistical programming and regulatory requirements. Modern clinical research demands not only accurate results but also standardized, reproducible, and regulatory-compliant datasets. This chapter reviews key literature related to oncology clinical trials, highlights the importance of CDISC standards, and explains the regulatory framework governing clinical data submissions.

2.2 KEYNOTE-024 STUDY REFERENCE

Protocol Reference (Design Inspiration) :

- The study design implemented in this project is inspired by the **KEYNOTE-024 clinical trial** (ClinicalTrials.gov Identifier: NCT02142738), which evaluated the efficacy of Pembrolizumab in patients with advanced non-small cell lung cancer (NSCLC).
- Selected elements from the original protocol such as study phase, treatment arms, population characteristics, and efficacy endpoints were adapted to construct a synthetic Phase III randomized clinical trial dataset.
- The purpose of using this reference was to ensure that the study design reflects real-world clinical trial standards while maintaining a fully synthetic and academic data environment.
- ⚠ Note: This project uses synthetic data and does not include any real patient-level data from the original clinical trial.

Product: MK-3475 (SCH-900475)
Protocol/Amendment No.: 024-11

1

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One Merck Drive
P.O. Box 100
Whitehouse Station, NJ 08889-0100, U.S.A.

Protocol-specific Sponsor Contact information can be found in the Investigator Trial File Binder (or equivalent).

TITLE:

A Randomized Open-Label Phase III Trial of Pembrolizumab versus Platinum based Chemotherapy in 1L Subjects with PD-L1 Strong Metastatic Non-Small Cell Lung Cancer

IND NUMBER: 116833

EudraCT NUMBER: 2014-000323-25

Figure 2.1: Reference Clinical Trial Protocol Design (KEYNOTE-024 Study)



Figure 2.2: Clinical Trial Registration Details from ClinicalTrials.gov (NCT02142738)

➤ The above snapshot represents the reference clinical trial design used for conceptual alignment.

Key elements adapted in this study include:

- Phase III randomized controlled trial design
- Two-arm structure (Pembrolizumab vs Chemotherapy/Placebo)
- Patient population: Advanced NSCLC
- Efficacy endpoints: Overall Survival (OS) and Progression-Free Survival (PFS)
-

(For comprehensive protocol documentation including clinical trial design, study parameters, and supporting reference materials, please refer to Digital Appendix - A: Protocol Reference)

2.3 CDISC STANDARDS IN CLINICAL RESEARCH

The Clinical Data Interchange Standards Consortium (CDISC) is a global, non-profit organization that develops standards to support the acquisition, exchange, submission, and analysis of clinical research data.

CDISC standards ensure that clinical trial data is structured in a consistent and regulatory-compliant manner. The two primary standards used in this study are:

2.3.1 Study Data Tabulation Model (SDTM)

SDTM is used for organizing and formatting raw clinical trial data into standardized domains. It provides a consistent structure for submission-ready datasets.

Key characteristics of SDTM:

- Organizes data into predefined domains such as Demographics (DM), Adverse Events (AE), Exposure (EX), Laboratory (LB), Vital Signs (VS), Tumor domains (TU, TR), Response Status (RS), and Disposition (DS)
- Ensures traceability from raw data to standardized datasets
- Supports regulatory review by providing structured data formats

2.3.2 Analysis Data Model (ADaM)

ADaM datasets are derived from SDTM and are specifically designed for statistical analysis.

Key characteristics of ADaM:

- Provides analysis-ready datasets
- Supports statistical procedures such as survival analysis and safety summaries
- Maintains traceability to SDTM datasets
- Includes datasets such as ADSL (subject-level), ADAE (adverse events), ADTTE (time-to-event), ADLB, ADVS, and ADRESP

2.4 REGULATORY AUTHORITIES AND GLOBAL COMPLIANCE

Clinical trial data must comply with strict regulatory guidelines before it can be accepted for drug approval. Various global regulatory authorities mandate the use of standardized data formats such as CDISC.

2.4.1 United States Food and Drug Administration (FDA)

The FDA requires submission of clinical trial data in CDISC-compliant formats, specifically SDTM and ADaM. The agency mandates electronic submissions using standardized datasets to ensure efficient review and validation.

2.4.2 European Medicines Agency (EMA)

The EMA supports the use of CDISC standards for clinical data submissions. It emphasizes data consistency, traceability, and transparency in safety and efficacy analysis.

2.5 “NO SAS = NO APPROVAL” CONCEPT

In modern clinical research, statistical programming using SAS has become a critical requirement for regulatory submissions.

Regulatory authorities do not accept unstructured or manually analyzed data (such as raw Excel files). Instead, they require validated, standardized datasets in CDISC format, typically generated using SAS.

The phrase “No SAS = No Approval” reflects the industry reality that:

- Clinical data must be transformed into SDTM and ADaM formats
- Statistical outputs (TLFs) must be generated using validated programming methods
- Data must be reproducible and traceable
- Submissions must comply with regulatory standards

Without SAS-based processing and CDISC-compliant datasets, a New Drug Application (NDA) is highly likely to be rejected by regulatory authorities.

2.6 DIGITAL APPENDIX – A : PROTOCOL REFERENCE

Due to the extensive details of the original clinical trial protocol and supporting documentation, complete reference materials are provided in the Digital Appendix.

▼ Content Available in Digital Appendix

- └ Reference Clinical Trial Protocol (KEYNOTE-024 Inspired Source)
- └ ClinicalTrials.gov Study Details (NCT02142738)
- └ Protocol Design Screenshots (Study Design, Arms, Population)
- └ Extracted Study Design Components used in this project
- └ Supporting notes for synthetic study adaptation
- └ Documentation for study structure alignment

▼ Digital Appendix Access Link

👉 [GOOGLE DRIVE LINK]

https://drive.google.com/drive/folders/1pWXV8X5fWu4bGOrt1N3E8AccQBriXPri?usp=drive_link



CHAPTER : 03

AIM, OBJECTIVES & HYPOTHESIS

3.1 AIM OF THE STUDY

The primary aim of this study is to design, implement, and validate an end-to-end clinical data processing and statistical programming framework using SAS software and CDISC standards. The study focuses on transforming raw clinical trial data into standardized SDTM datasets, deriving analysis-ready ADaM datasets, and generating validated Tables, Listings, and Figures (TLFs).

This project utilizes synthetic oncology clinical trial data to simulate real-world scenarios and demonstrate industry-level statistical programming workflows. The emphasis is on ensuring data standardization, traceability, and reproducibility of results using validated SAS programming techniques.

3.2 OBJECTIVES OF THE STUDY

The study is structured into multiple objectives representing different stages of the clinical data lifecycle:

3.2.1 Primary Objectives

- To develop a structured workflow for transforming raw clinical data into CDISC-compliant datasets
- To implement SDTM mapping for standardizing clinical trial data across multiple domains
- To derive ADaM datasets for statistical analysis and reporting
- To generate Tables, Listings, and Figures (TLFs) using SAS procedures
- To perform Quality Control (QC) validation using independent programming techniques

3.2.2 Secondary Objectives

- To simulate a Phase III oncology clinical trial dataset using synthetic data
- To demonstrate traceability from raw data to final statistical outputs
- To implement time-to-event analysis (e.g., survival analysis) using SAS procedures such as PROC LIFETEST
- To develop a structured and reproducible SAS programming workflow
- To prepare outputs aligned with Clinical Study Report (CSR) requirements

3.2.3 Technical Objectives (Industry-Oriented)

- To replicate real-world clinical SAS programming practices
- To apply CDISC standards (SDTM and ADaM) in a structured manner

3.3 RESEARCH HYPOTHESIS

Since the study utilizes synthetic data and focuses on statistical programming validation rather than clinical inference, the hypothesis is defined in terms of the SAS-based data processing framework.

NULL HYPOTHESIS (H_0) :

The SAS-based statistical programming framework fails to correctly transform raw clinical data into standardized SDTM and ADaM datasets and does not generate accurate TLF outputs.

ALTERNATIVE HYPOTHESIS (H_1) :

The SAS-based statistical programming framework successfully transforms raw clinical data into CDISC-compliant SDTM and ADaM datasets and generates accurate, reproducible, and validated TLF outputs.

3.4 HYPOTHESIS JUSTIFICATION

In traditional clinical research, hypotheses are typically defined to evaluate the efficacy and safety of a drug or intervention. However, in the present study, the primary focus is not clinical outcome evaluation but the validation of a statistical programming workflow.

Due to the use of synthetic data, the results generated are not intended for real clinical decision-making. Instead, the study aims to demonstrate that the SAS programming pipeline is capable of:

- Accurately transforming raw data into standardized formats
- Maintaining traceability across different stages of data processing
- Generating consistent statistical outputs
- Ensuring quality through independent validation

Therefore, the hypothesis is defined in terms of the successful implementation and validation of the SAS-based clinical data workflow.

CHAPTER : 04

STUDY DESIGN & METHODOLOGY

4.1 STUDY DESIGN

4.1.1 INTRODUCTION

This study is designed as a synthetic simulation of a Phase III oncology clinical trial framework to support SAS-based statistical programming and CDISC implementation. The study design is adapted from established clinical trial protocols to ensure that the generated datasets reflect real-world clinical data structures.

It is important to note that this study does not involve actual patient recruitment or clinical intervention. Instead, it utilizes a simulated dataset to demonstrate the end-to-end data processing pipeline used in clinical research.

4.1.2 PROTOCOL COMPARISON (REFERENCE VS SIMULATED STUDY)

To ensure realism in the synthetic dataset, the study design was inspired by the KEYNOTE-024 clinical trial, which evaluated pembrolizumab in advanced NSCLC patients.

PARAMETER	KEYNOTE-024 (REFERENCE STUDY)	ONCO-PEMBRO-500 (SIMULATED STUDY)
Study Phase	Phase III	Phase III (Simulated)
Study Type	Randomized, Open-label	Randomized, Double Blind
Population	Advanced NSCLC	Stage IV NSCLC (Synthetic)
Sample Size	~305 subjects	500 subjects
Treatment Arm	Pembrolizumab Chemotherapy	vs Pembrolizumab + Chemo vs Placebo + Chemo
Endpoints	OS, PFS	OS, PFS, BOR (Simulated)

[Table 4.1: Protocol Comparison Between KEYNOTE-024 (Reference Study) and ONCO-PEMBRO-500 (Simulated Study)]

The simulated study design was developed to closely mimic real clinical trial conditions while allowing flexibility for synthetic data generation and statistical programming.

4.1.3 STUDY DESIGN

The study follows a structured Phase III clinical trial framework with the following characteristics:

- **Study Phase:** Phase III (Simulated Model)
- **Study Type:** Randomized, Double Blind, Parallel Group
- **Study Design:** Two-arm comparative study
- **Randomization Ratio:** 1:1 allocation

The double-blind design ensures that both subjects and investigators are unaware of treatment assignments, thereby minimizing bias in outcome evaluation (simulated context).

4.1.4 STUDY POPULATION AND SAMPLE SIZE

The study population consists of synthetic patient-level data representing individuals diagnosed with Stage IV Non-Small Cell Lung Cancer (NSCLC).

- **Total Sample Size:** 500 subjects
- **Arm A:** 250 subjects
- **Arm B:** 250 subjects

The sample size is evenly distributed between treatment arms to ensure balanced comparison and consistency in statistical analysis.

4.1.5 TREATMENT ARMS

The study includes two treatment groups designed to replicate real-world oncology trial conditions:

Arm A (Experimental Group)

- Pembrolizumab 200 mg administered intravenously every 3 weeks (Q3W)
- Combined with standard platinum-based chemotherapy:
 - Carboplatin
 - Pemetrexed

Arm B (Control Group)

- Placebo administered intravenously every 3 weeks (Q3W)
- Combined with the same chemotherapy regimen:
 - Carboplatin
 - Pemetrexed

4.1.6 STUDY ENDPOINTS (OVERVIEW)

The study includes key oncology endpoints for simulation purposes:

- **Overall Survival (OS)**
- **Progression-Free Survival (PFS)**
- **Best Overall Response (BOR)**

These endpoints are used to demonstrate statistical analysis techniques and TLF generation within the SAS programming framework.

4.2 RESEARCH METHODOLOGY

4.2.1 INTRODUCTION TO RESEARCH METHODOLOGY

The research methodology adopted in this study follows a structured end-to-end statistical programming workflow used in modern oncology clinical trials. The methodology was designed to simulate real-world clinical trial data processing standards using SAS OnDemand® and CDISC implementation frameworks.

The complete workflow includes synthetic raw data preparation, SAS-based data import, SDTM mapping, ADaM dataset derivation, TLF generation, and independent Quality Control (QC) validation. Each stage of the methodology was developed to maintain traceability, reproducibility, regulatory alignment, and standardized statistical reporting practices.

The overall methodology emphasizes practical implementation of CDISC standards within a simulated oncology clinical trial environment.

4.2.2 RAW DATA ACQUISITION & CDM EXPORT

Synthetic raw datasets were created to simulate Clinical Data Management (CDM) export files commonly received in real clinical trials. The datasets were prepared in CSV format and designed to represent operational clinical data collected during oncology studies.

Multiple domain-specific raw datasets were generated, including:

- Demographics (DM)
- Adverse Events (AE)
- Exposure (EX)
- Laboratory Data (LB)
- Vital Signs (VS)
- Tumor Assessments (TU/TR/RS)
- Disposition (DS)

The raw datasets were structured to resemble real-world clinical trial data complexity and variability while maintaining a fully synthetic and simulated environment.

4.2.3 SAS DATA IMPORT & PRE-SDTM PROCESSING

The raw CSV datasets were imported into SAS using PROC IMPORT procedures within SAS OnDemand®. After import, initial preprocessing activities were performed to prepare the datasets for SDTM implementation.

Preprocessing activities included:

- Variable formatting and standardization
- Date conversion into ISO 8601 format
- Missing value handling
- Label assignment and metadata preparation
- Dataset structure verification
- Variable length and type validation

These preprocessing steps ensured consistency and compatibility with downstream CDISC implementation workflows.

4.2.4 SDTM MAPPING & STANDARDIZATION

The imported raw datasets were transformed into CDISC Study Data Tabulation Model (SDTM) domains using predefined mapping specifications and SAS programming logic.

Standard SDTM implementation principles were followed, including:

- Domain-level dataset standardization
- Controlled terminology implementation
- Variable derivation logic
- Identifier assignment (STUDYID, USUBJID, etc.)
- Supplemental qualifier handling using SUPP domains
- Traceability maintenance between raw and SDTM datasets

A total of 14 SDTM domains were implemented during the study, including core domains and supplemental datasets.

4.2.5 ADAM DATASET DERIVATION

Analysis Data Model (ADaM) datasets were derived from SDTM domains to create analysis-ready datasets suitable for statistical reporting and TLF generation.

The derivation process included:

- Analysis population flag generation
- Treatment arm assignment
- Baseline variable derivation
- Analysis visit calculations
- Time-to-event parameter derivation
- Response endpoint derivation

Key ADaM datasets implemented in this study included:

- ADSL – Subject-Level Analysis Dataset
- ADAE – Adverse Event Analysis Dataset
- ADLB – Laboratory Analysis Dataset
- ADVS – Vital Signs Analysis Dataset
- ADRESP – Tumor Response Dataset
- ADTTE – Time-to-Event Analysis Dataset

These datasets formed the foundation for downstream statistical analysis and output generation.

4.2.6 TLF GENERATION (TABLES, LISTINGS & FIGURES)

Statistical outputs including Tables, Listings, and Figures (TLFs) were generated using validated ADaM datasets through SAS programming procedures.

The TLF outputs were designed according to standard clinical reporting practices and included:

- Demographic summary tables
- Treatment exposure summaries
- Adverse event summaries
- Laboratory analysis tables
- Subject-level listings
- Kaplan–Meier survival plots
- Tumor response analysis outputs

Outputs were generated in RTF-compatible formats suitable for Clinical Study Report (CSR) integration and regulatory documentation workflows.

4.2.7 QUALITY CONTROL & VALIDATION STRATEGY

Independent Double Programming (IDP) methodology was implemented to perform Quality Control (QC) validation of generated outputs.

Under this approach:

- Production programs generated the primary outputs
- Separate QC programs were independently developed
- Both outputs were compared for consistency and accuracy
- Discrepancies were investigated and resolved

Validation activities included:

- Subject count verification
- Summary statistic comparison
- Dataset structure validation
- Listing record verification
- Graphical output confirmation

The QC methodology ensured reproducibility, reliability, and validation integrity across all statistical outputs.

4.2.8 REGULATORY SUBMISSION READINESS

Final validated outputs were generated in standardized formats compatible with regulatory submission workflows and Clinical Study Report (CSR) integration.

The project workflow maintained:

- CDISC-compliant dataset structures
- Traceability between raw data and outputs
- Standardized output formatting
- Validation documentation support
- Regulatory-oriented reporting practices

The final deliverables demonstrate a submission-oriented statistical programming workflow aligned with industry-standard clinical data processing environments.

CHAPTER : 05

WHY SYNTHETIC DATA?

5.1 INTRODUCTION

In clinical research, access to real patient-level data is highly restricted due to strict privacy regulations and ethical considerations. As a result, the use of synthetic data has become an effective alternative for academic research, technical validation, and training in clinical data analysis.

This study utilizes synthetic clinical trial data to simulate real-world oncology datasets while maintaining compliance with data privacy and regulatory standards.

5.2 DATA PRIVACY AND REGULATORY COMPLIANCE

Clinical trial data contains sensitive patient information, which is protected under global data privacy regulations such as:

- Health Insurance Portability and Accountability Act (HIPAA)
- General Data Protection Regulation (GDPR)

These regulations impose strict controls on the use, sharing, and processing of patient data. Access to real clinical datasets is limited and often requires regulatory approvals, data use agreements, and ethical clearance.

Synthetic data eliminates these challenges by generating artificial datasets that mimic real clinical data without containing any identifiable patient information. This ensures that the study remains compliant with privacy laws while allowing full access to the data for analysis.

5.3 TECHNICAL VALIDATION OF SAS PROGRAMMING

The primary objective of this study is to validate the SAS-based statistical programming pipeline rather than to evaluate clinical outcomes.

Synthetic data provides a controlled environment where:

- Data structures can be predefined
- Variables can be systematically generated
- Scenarios can be simulated for testing

This enables accurate validation of data transformation processes including SDTM mapping, ADaM derivation, and TLF generation.

CHAPTER : 06

DATA DESIGN & RAW DATA PREPARATION

6.1 INTRODUCTION

This chapter describes the design and preparation of raw clinical trial data prior to SDTM implementation. It includes the mapping strategy, domain-wise data structuring, and synthetic data generation approach used to simulate CDM export datasets.

The objective of this stage is to ensure that raw data is well-structured, standardized, and suitable for downstream SDTM and ADaM transformations using SAS programming.

6.2 MAPPING SHEET DESIGN

Mapping is the process of transforming raw clinical trial data into standardized SDTM domains by defining relationships between raw variables and CDISC-compliant variables.

A structured mapping strategy was developed to ensure consistency, traceability, and standardization across all datasets. Each raw variable was mapped to its corresponding SDTM variable based on domain specifications.

The following domains were considered for mapping:

- DM – Demographics
 - AE – Adverse Events
 - EX – Exposure
 - LB – Laboratory Tests
 - VS – Vital Signs
 - TU – Tumor Identification
 - TR – Tumor Response
 - RS – Response Status
 - DS – Disposition
-

6.3 REPRESENTATIVE MAPPING TABLES

The following tables present representative examples of domain-level mapping used in this study.

TABLE 1 — CORE DM VARIABLES MAPPING

(1) DOMAIN : DM (DEMOGRAPHICS)

<u>CRF Field</u>	<u>RAW Variable</u>	<u>SDTM Variable</u>	<u>Description</u>	<u>Derivation Logic</u>	<u>Controlled Terminology</u>	<u>Notes</u>
Study ID	STUDYID	STUDYID	Study Identifier	Direct	ONCO-PEMBRO-500	Constant
Site ID	SITEID	SITEID	Site Identifier	Direct	Alphanumeric	—
Subject ID	SUBJID	SUBJID	Subject ID assigned by site	Direct	001–500	Unique within site
—	—	USUBJID	Unique Subject Identifier	Concatenate: STUDYID-SITEID-SUBJID	—	Key variable
Screening Date	SCRNDT	—	Screening date	Not included in SDTM DM	—	Used for audit only
Randomization Date	RANDDT	RFSTDTC	Reference Start Date	Convert to ISO 8601 (YYYY-MM-DD)	ISO 8601	Required for ADTTE
Randomization Arm	RAND_ARM	ARM	Description of treatment arm	Direct	"Arm A" / "Arm B"	From CRF
Randomization Arm	RAND_ARM	ARMCD	Short coded treatment arm	If ARM="Arm A" → A; If "Arm B" → B	A / B	CDISC requirement
Age	AGE	AGE	Age in years	Direct	Numeric	AGEU always = YEARS
—	—	AGEU	Unit for age	Set as "YEARS"	YEARS	Standard
Sex	SEX	SEX	Biological sex	Direct	M / F	SDTM CT
Race	RACE	RACE	Race category	Direct	Asian / White / Black	SDTM CT
Ethnicity	ETHNIC	ETHNIC	Optional	Direct	Hispanic / Non-Hispanic	Optional
Country	COUNTRY	COUNTRY	Enrollment Country	Direct	INDIA / USA / EU	—
ECOG Status	ECOG	ECOG	Baseline Performance Status	Direct	0 / 1	Oncology indicator
Smoking Status	SMK	— (Goes to SUPPDM)	Smoking history	Direct	Yes / No	→ SUPPDM

Height (cm)	HEIGHT	— (SUPPDM.QVAL)	Height	Direct	Numeric	Supplemental
Weight (kg)	WEIGHT	— (SUPPDM.QVAL)	Weight	Direct	Numeric	Supplemental

[Table 6.1 : Core DM Variables Mapping – CRF to SDTM (Demographics Domain)]

TABLE 2 — SUPPDM MAPPING (HEIGHT, WEIGHT, SMOKING)

(Non-standard qualifiers for DM)

<u>CRF Field</u>	<u>RAW Variable</u>	<u>SUPPDM Variable</u>	<u>Value</u>	<u>Explanation</u>
Smoking Status	SMK	QNAM SMOKER	= QVAL Yes/No	= Baseline smoking history
Height (cm)	HEIGHT	QNAM HEIGHT	= QVAL numeric	= Supplemental variable
Weight (kg)	WEIGHT	QNAM WEIGHT	= QVAL numeric	= Supplemental variable

Table 6.2: SUPPDM Mapping – Non-Standard Qualifiers (Height, Weight, Smoking Status)

▼ **Important Note**

The above tables represent key domain-level mapping structures used for SDTM implementation. Complete mapping sheets for all domains, including LB, EX, VS, TU, TR, RS, and DS, are provided in the Digital Appendix (Appendix B) for detailed reference.

[Detailed mapping sheets for all SDTM domains, including complete SUPPQUAL structures, are provided in Digital Appendix – B for comprehensive reference and validation.]

6.4 RAW DATA CREATION (CDM EXPORT SIMULATION)

Raw datasets were designed to simulate a Clinical Data Management (CDM) export structure using CSV format. These datasets represent subject-level and event-level clinical data collected during a clinical trial.

The data was generated based on the study design and protocol framework to closely resemble real-world oncology clinical trial datasets.

6.4.1 Domain-wise Raw Data Structure

A total of nine raw datasets were created, each corresponding to a specific clinical domain:

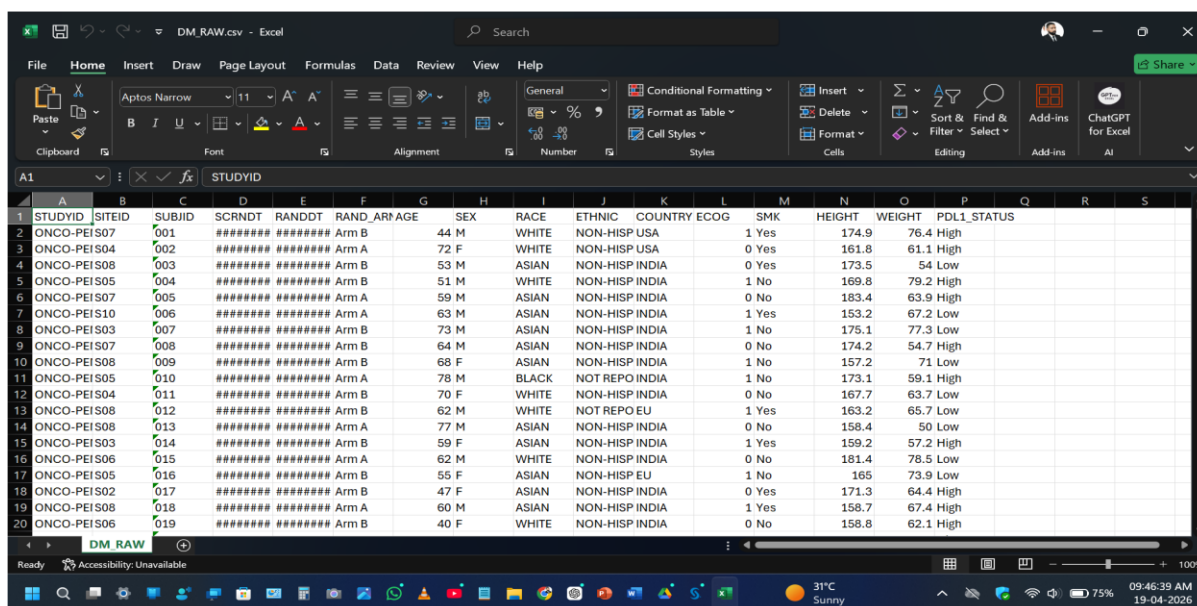
- DM – Demographics
- EX – Expo
- VS – Vital Signs
- TR – Tumor Response
- DS – Disposition
- AE – Adverse Events
- LB – Laboratory Tests
- TU – Tumor Identification
- RS – Response Status

Each dataset was structured in tabular format with variables aligned to support SDTM mapping and downstream analysis.

6.4.2 Representative Raw Data Samples

To illustrate the structure of raw datasets, representative samples from key domains are presented below.

- 1. DM Raw Dataset (CSV Format) – Excel View**



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
	STUDYID	SITEID	SUBJID	SCNRND	RANDDT	RAND_ARM	AGE	SEX	RACE	ETHNIC	COUNTRY	ECOG	SMK	HEIGHT	WEIGHT	PDL1_STATUS			
1	ONCO-PEIS07	'001	*****	*****	*****	Arm B	44	M	WHITE	NON-HISP USA			1 Yes	174.9	76.4	High			
2	ONCO-PEIS04	'002	*****	*****	*****	Arm A	72	F	WHITE	NON-HISP USA			0 Yes	161.8	61.1	High			
3	ONCO-PEIS08	'003	*****	*****	*****	Arm B	53	M	ASIAN	NON-HISP INDIA			0 Yes	173.5	54	Low			
4	ONCO-PEIS05	'004	*****	*****	*****	Arm B	51	M	WHITE	NON-HISP INDIA			1 No	169.8	79.2	High			
5	ONCO-PEIS07	'005	*****	*****	*****	Arm A	59	M	ASIAN	NON-HISP INDIA			0 No	183.4	63.9	High			
6	ONCO-PEIS10	'006	*****	*****	*****	Arm A	63	M	ASIAN	NON-HISP INDIA			1 Yes	153.2	67.2	Low			
7	ONCO-PEIS03	'007	*****	*****	*****	Arm B	73	M	ASIAN	NON-HISP INDIA			1 No	175.1	77.3	Low			
8	ONCO-PEIS07	'008	*****	*****	*****	Arm B	64	M	ASIAN	NON-HISP INDIA			0 No	174.2	54.7	High			
9	ONCO-PEIS08	'009	*****	*****	*****	Arm B	68	F	ASIAN	NON-HISP INDIA			1 No	157.2	71	Low			
10	ONCO-PEIS05	'010	*****	*****	*****	Arm A	78	M	BLACK	NOT REPO INDIA			1 No	173.1	59.1	High			
11	ONCO-PEIS04	'011	*****	*****	*****	Arm B	70	F	WHITE	NON-HISP INDIA			0 No	167.7	63.7	Low			
12	ONCO-PEIS08	'012	*****	*****	*****	Arm B	62	M	WHITE	NOT REPO EU			1 Yes	163.2	65.7	Low			
13	ONCO-PEIS08	'013	*****	*****	*****	Arm A	77	M	ASIAN	NON-HISP INDIA			0 No	158.4	50	Low			
14	ONCO-PEIS03	'014	*****	*****	*****	Arm B	59	F	ASIAN	NON-HISP INDIA			1 Yes	159.2	57.2	High			
15	ONCO-PEIS06	'015	*****	*****	*****	Arm A	62	M	WHITE	NON-HISP INDIA			0 No	181.4	78.5	Low			
16	ONCO-PEIS05	'016	*****	*****	*****	Arm B	55	F	ASIAN	NON-HISP EU			1 No	165	73.9	Low			
17	ONCO-PEIS02	'017	*****	*****	*****	Arm B	47	F	ASIAN	NON-HISP INDIA			0 Yes	171.3	64.4	High			
18	ONCO-PEIS08	'018	*****	*****	*****	Arm A	60	M	ASIAN	NON-HISP INDIA			1 Yes	158.7	67.4	High			
19	ONCO-PEIS06	'019	*****	*****	*****	Arm B	40	F	WHITE	NON-HISP INDIA			0 No	158.8	62.1	High			

[Figure 6.1 : DM Raw Dataset (CSV Format) – Excel View]

2. AE Raw Dataset (CSV Format) : [EXCEL/CSV SCREENSHOT]

[Figure 6.2: Raw AE Dataset (CSV Format) – Sample Adverse Event Data in Excel]

[Complete domain-wise raw datasets (CDM export files), including DM, AE, EX, LB, VS, TU, TR, and related structures, are provided in Digital Appendix – C for full data transparency and reference.]

6.4.3 : Sas Output Screenshot After Import In Sas Studio

[Figure 6.3: SDTM Dataset View in SAS Studio – Post Import Verification]

6.4.4 Raw Data Design Characteristics

The raw datasets were developed with the following characteristics to simulate realistic clinical trial data:

- Subject-level identifiers (e.g., USUBJID) maintained across domains
- Consistent variable naming conventions
- Logical relationships between datasets
- Balanced treatment arm distribution
- Event-based and longitudinal data representation

These characteristics ensure compatibility with SDTM standards and enable accurate transformation during the data standardization process.

6.5 : DIGITAL APPENDIX - B : DOMAIN MAPPING SHEET

▼ Content Available in Digital Appendix

1. DM – Demographics – MAPPING SHEET , 02. AE - Adverse Events – MAPPING SHEET
03. EX – Exposure – MAPPING SHEET , 04. LB - Laboratory Tests – MAPPING SHEET
05. VS - Vital Signs – MAPPING SHEET , 06. TU - Tumor Identification – MAPPING SHEET
07. TR - Tumor Response – MAPPING SHEET ,08. RS - Response Status – MAP SHEET
09. DS – Disposition – MAPPING SHEET

- **Digital Appendix Access Link**

https://drive.google.com/drive/folders/1xmuZJJpVpR504W7geSZt55lXL0U-DpNI?usp=drive_link



6.6 : DIGITAL APPENDIX - C : RAW DATA (CDM EXPORT FILES)

Due to the extensive size of raw datasets, complete domain-wise raw data files for all nine domains are provided in the Digital Appendix (Appendix C) for detailed reference.

▼ Content Available in Digital Appendix

- ✓ B1: DM Raw Data, B2: AE Raw Data , B3: EX Raw Data
- ✓ B4: LB Raw Data
- ✓ B5: VS Raw Data
- ✓ B6: TU Raw Data
- ✓ B7: TR Raw Data
- ✓ B8: RS Raw Data
- ✓ **B9: DS Raw Data**

- Digital Appendix Access Link

👉 [GOOGLE DRIVE LINK]

https://drive.google.com/drive/folders/1jtsdULaJ4E--jn3mlbGBcMZ6UY6IDhxl?usp=drive_link



CHAPTER : 07

SDTM

IMPLEMENTATIO

7.1 INTRODUCTION

The Study Data Tabulation Model (SDTM) is a standard developed by CDISC for organizing and formatting clinical trial data for regulatory submission. It provides a structured framework to convert raw datasets into standardized domains.

In this study, raw clinical datasets were transformed into SDTM-compliant domains using SAS programming. This process ensures consistency, traceability, and regulatory compliance of clinical data.

7.2 SDTM OVERVIEW

SDTM organizes clinical data into predefined domains, where each domain represents a specific type of clinical information such as demographics, adverse events, laboratory results, and tumor assessments.

Each SDTM dataset follows a standardized structure with:

- Domain-specific variables
- Controlled terminology
- Consistent naming conventions
- One record per observation

The implementation of SDTM allows regulatory reviewers to easily understand and analyze the data regardless of study origin.

7.3 SDTM DOMAINS IMPLEMENTED

In this study, a total of fourteen SDTM datasets were created, including parent and supplemental domains.

◆ Parent Domains

DM – Demographics	, AE – Adverse Events	EX – Exposure
LB – Laboratory Tests	, VS – Vital Signs	TU – Tumor Identification
Tumor Response	, RS – Response Status	TR – Tumor Response
		DS – Disposition

◆ Supplemental Domains (SUPPQUAL)

Supplemental domains are used to store non-standard variables that are not part of the standard SDTM structure.

SUPPDM , SUPPAE , SUPPEX , SUPPLB , SUPPVS

These domains ensure that additional information is retained without violating SDTM standards.

7.4 SDTM Dataset Implementation Creation Using SAS : (Domain-wise Demonstration)

Each SDTM domain was developed using SAS programming, and a standardized implementation approach was followed. A summarized view of each domain, including code, log, and output, is presented below.”

The general workflow followed was:

- Import raw data using PROC IMPORT
- Apply data transformation using DATA step
- Assign SDTM variables and labels
- Export final SDTM datasets

7.5.1 ♦ DM – Demographics Domain

The DM domain contains subject-level demographic information such as age, sex, and treatment arm. The dataset was generated using SAS DATA step and validated for SDTM compliance.

The screenshot displays the SAS Studio interface. The top pane shows the SAS code for creating the SDTM DM dataset. The code includes comments, library assignments, data import, and sorting steps.

```
1 /*****  
2 SDTM DM CREATION PROGRAM (FINAL)  
3 Study: ONCO-PEMBRO-500  
4 Domain: DM (Demographics)  
5 *****/  
6  
7 /*  
8 STEP 1: Assign Libraries  
9 *****/  
10 libname RAW "/home/u64086073/01_RAW_DATA";  
11 libname SDTM "/home/u64086073/03_SDTH";  
12  
13 /*  
14 STEP 2: Import RAW DM Dataset  
15 *****/  
16 proc import datafile="/home/u64086073/01_RAW_DATA/DM_RAW.csv"  
17 out=raw_dm  
18 dds=csv  
19 replace;  
20 guessingrows=500;  
21 run;  
22  
23 /*  
24 STEP 3: Sort RAW Data  
25 *****/  
26  
27 proc sort data=raw_dm;  
28 by SITEID SUBJID;  
29 run;  
30  
31  
32  
33 /***/
```

The bottom pane shows the log and output verification. The log displays the execution of the code, and the output shows the contents of the SDTM DM dataset.

Obs	STUDYID	DOMAIN	USUBJID	SITEID	SUBJID	ARMCD	AGE	AGEU	SEX	RACE	ETHNIC	COUNTRY	ECOU	RDSTDTC	SCRNRT	RANRDT	RANDAT	RAND_ARM	SMK	HEIGHT	WEIGHT	POLY_STATUS
1	ONCO-PEMBRO-500	DM	ONCO-PEMBRO-500-801-01	801	22	A	71	YEAR	F	WHITE	NON-HISPANIC	INDIA	1	23403	2024-01-12	2024-01-12	2024-01-12	Arm A	Yes	161.7	67.3	High
2	ONCO-PEMBRO-500	DM	ONCO-PEMBRO-500-801-02	801	28	A	72	YEAR	M	ASIAN	NOT REPORTED	INDIA	1	23484	2024-02-13	2024-02-13	2024-02-13	Arm A	No	168.2	49.7	Low
3	ONCO-PEMBRO-500	DM	ONCO-PEMBRO-500-801-03	801	52	B	45	YEAR	M	WHITE	NON-HISPANIC	INDIA	1	23484	2024-02-13	2024-02-13	2024-02-13	Arm B	No	168.7	65	High
4	ONCO-PEMBRO-500	DM	ONCO-PEMBRO-500-801-04	801	76	A	46	YEAR	M	BLACK	NON-HISPANIC	INDIA	0	23481	2024-01-11	2024-01-11	2024-01-11	Arm A	No	170.8	63	High
5	ONCO-PEMBRO-500	DM	ONCO-PEMBRO-500-801-05	801	84	B	51	YEAR	F	ASIAN	NON-HISPANIC	INDIA	1	23480	2024-02-13	2024-02-13	2024-02-13	Arm B	No	172	63.4	Low
6	ONCO-PEMBRO-500	DM	ONCO-PEMBRO-500-801-06	801	86	A	56	YEAR	M	ASIAN	NON-HISPANIC	INDIA	0	23415	2024-01-23	2024-01-23	2024-01-23	Arm A	Yes	166.6	66.9	Low
7	ONCO-PEMBRO-500	DM	ONCO-PEMBRO-500-801-07	801	87	B	71	YEAR	M	WHITE	NON-HISPANIC	INDIA	1	23480	2024-01-28	2024-01-28	2024-01-28	Arm B	Yes	164.3	61.6	High
8	ONCO-PEMBRO-500	DM	ONCO-PEMBRO-500-801-08	801	102	B	46	YEAR	F	WHITE	NON-HISPANIC	INDIA	0	23435	2024-02-26	2024-02-26	2024-02-26	Arm B	No	164.3	64.4	Low
9	ONCO-PEMBRO-500	DM	ONCO-PEMBRO-500-801-101	801	111	A	65	YEAR	F	ASIAN	NON-HISPANIC	INDIA	0	23436	2024-02-26	2024-02-26	2024-02-26	Arm A	Yes	172.4	34.9	High
10	ONCO-PEMBRO-500	DM	ONCO-PEMBRO-500-801-115	801	115	A	73	YEAR	M	ASIAN	NON-HISPANIC	INDIA	1	23401	2024-01-18	2024-01-18	2024-01-18	Arm A	No	157.7	95.1	High

The bottom pane also shows the contents of the SDTM DM dataset, including the number of observations (500) and variables (21).

[Figure 7.1: SDTM DM Domain Implementation – SAS Code, Log, and Output Verification]

7.5 SDTM DATASET VERIFICATION

All SDTM datasets were verified for:

- Variable consistency , Proper labeling , Domain structure , Data accuracy

The datasets were successfully generated and validated using SAS. To verify the successful creation of all SDTM datasets, a consolidated SAS library view is presented below.

👉 [SAS LIBRARY SCREENSHOT SHOWING ALL 14 DATASETS]

STUDYID	DOM	USUBJID	ARM	AG	SITEID	SUBJID	SCRNDT
1	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-22 A	YEARS 501			22	2024-01-12 2i
2	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-26 A	YEARS 501			26	2024-02-13 2i
3	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-52 B	YEARS 501			52	2024-02-18 2i
4	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-79 A	YEARS 501			79	2024-01-11 2i
5	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-84 B	YEARS 501			84	2024-02-27 2i
6	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-88 A	YEARS 501			88	2024-01-23 2i
7	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-92 B	YEARS 501			92	2024-01-28 2i
8	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-10 B	YEARS 501			102	2024-02-25 2i
9	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-11 A	YEARS 501			111	2024-02-26 2i
10	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-11 A	YEARS 501			115	2024-01-18 2i
11	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-12 B	YEARS 501			125	2024-02-12 2i
12	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-14 B	YEARS 501			145	2024-02-07 2i
13	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-14 B	YEARS 501			146	2024-02-26 2i
14	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-15 B	YEARS 501			157	2024-01-03 2i
15	ONCO-PEMBRO-SI DM	ONCO-PEMBRO-500-S01-16 B	YEARS 501			165	2024-02-04 2i

Figure 7.2: Consolidated SAS Library View – Verification Of All SDTM Domain Datasets

7.6 DIGITAL APPENDIX - D : SDTM DATASETS

Due to the large volume of SDTM datasets and supporting files, complete documentation is provided in the Digital Appendix.

▼ **Digital Appendix Access Link** 👉 [GOOGLE DRIVE LINK]

https://drive.google.com/drive/folders/1Fmvz2Xon0X-oRqkYa2_wRGKhI4HbMs-s?usp=drive_link



CHAPTER : 08

ADaM ANALYSIS

DESIGN &

DERIVATION

◆ 8.0 INTRODUCTION

This chapter describes the design and derivation of Analysis Data Model (ADaM) datasets used for statistical analysis. ADaM datasets are derived from SDTM datasets and are structured to support analysis, reporting, and generation of Tables, Listings, and Figures (TLFs).

The focus of this chapter is to present dataset-level design, key derivation logic, and traceability from SDTM to ADaM.

◆ 8.1 ADAM OVERVIEW

ADaM (Analysis Data Model) is a CDISC standard used to create analysis-ready datasets. These datasets are specifically structured to support statistical procedures and reporting.

Key characteristics of ADaM datasets include:

- Analysis-ready structure
 - Clear traceability to SDTM datasets
 - Derived variables for statistical analysis
 - Support for reproducible results
-

◆ 8.2 ADAM DATASETS IMPLEMENTED

The following ADaM datasets were created in this study:

- ADSL – Subject-Level Analysis Dataset
 - ADAE – Adverse Events Analysis Dataset
 - ADLB – Laboratory Analysis Dataset
 - ADVS – Vital Signs Analysis Dataset
 - ADTTE – Time-to-Event Analysis Dataset
 - ADRESP – Response Analysis Dataset
-

◆ 8.3 ADAM ANALYSIS DESIGN

ADAM IMPLEMENTATION / DERIVATION (ANALYSIS READY)		
<i>Transforming SDTM Domains into ADaM IG v1.1 Standards for Statistical Analysis</i>		
What is ADaM?: ADaM datasets are specifically designed for Statistical Analysis.		
THE 6 GENERATED ADAM DATASETS (TRACEABILITY MATRIX)		
ADaM (Output)	Description (Use Case)	Source SDTM Domains (Input)
1. ADSL	Subject Level Analysis (Demographics, Flags)	DM + DS + EX
2. ADAE	Adverse Events Analysis (Safety)	AE + ADSL
3. ADLB	Laboratory Analysis (Safety - Toxicity Grades)	LB + ADSL
4. ADVS	Vital Signs Analysis (Safety)	VS + ADSL
5. ADRESP	Tumor Response Analysis (Efficacy - BOR)	ADSL + RS
6. ADTTE	Time-to-Event Analysis (Efficacy - Survival)	ADSL + ADVS

🔥 ◆ 8.4 DERIVATION SPECIFICATION (MAIN PART)

◆ 8.4.1 ADSL Derivation (FULL DETAIL)

The ADSL dataset is the subject-level dataset and serves as the foundation for all analysis datasets. It contains one record per subject and includes treatment assignment and demographic variables.

PURPOSE

Provides one record per subject and serves as the primary merge backbone for all other ADaM datasets.

◆ SOURCE DATASETS

- SDTM.DM – Demographics
- SDTM.EX – Exposure
- SDTM.DS - Disposition

◆ KEY VARIABLES DERIVED

- STUDYID
- USUBJID
- AGE, SEX, RACE
- TRT01P / TRT01A (planned & actual treatment)
- TRTSDT / TRTEDT (treatment start/end date)
- SAFFL, ITTFL (analysis flags)

◆ MERGE LOGIC

Primary key: USUBJID

DM → base subject list

EX → merged by USUBJID to derive treatment dates and flags

A. IDENTIFIER VARIABLES

Variable	Label	Source (sdm)	Merge logic	Derivation logic
STUDYID	Study Identifier	DM	DM base	Direct
USUBJID	Unique Subject Identifier	DM	DM base	Direct
SITEID	Site Identifier	DM	DM base	Direct
SUBJID	Subject Identifier	DM	DM base	Direct

B. TREATMENT VARIABLES

Variable	Label	Source (sdm)	Merge logic	Derivation logic
ARM	Planned Treatment Arm	DM	DM base	Direct
ARMCD	Planned Arm Code	DM	DM base	Direct
TRT01P	Planned Treatment	DM	DM base	ARM
TRT01PN	Planned Treatment (Numeric)	DM	DM base	Arm A=1, Arm B=2
TRT01A	Actual Treatment	EX	DM + EX by USUBJID	First exposure

C. POPULATION FLAGS

Variable	Label	Source (SDTM)	Merge Logic	Derivation Logic
ITTFL	Intent-to-Treat Population Flag	DM	DM base	Y if randomized
SAFFL	Safety Population Flag	EX	DM + EX	Y if ≥ 1 dose
EFFFL	Efficacy Population Flag	RS	ADSL + RS	Y if post-baseline assessment
COMPFL	Completed Study Flag	DS	ADSL + DS	Y if DSDECOD=COMPLETED

D. KEY DATES & STUDY DAYS

Variable	Label	Source (SDTM)	Merge Logic	Derivation Logic
RFSTDTC	Reference Start Date	DM	DM base	Direct
RANDDT	Randomization Date	DM	DM base	RFSTDTC
TRTSDT	Treatment Start Date	EX	DM + EX	Min(EXSTDTC)

E. DEMOGRAPHICS

Variable	Label	Source (SDTM)	Merge Logic	Derivation Logic
AGE	Age	DM	DM base	Direct
AGEU	Age Units	DM	DM base	Direct
SEX	Sex	DM	DM base	Direct
RACE	Race	DM	DM base	Direct
COUNTRY	Country	DM	DM base	Direct
ECOG	ECOG Status	DM	DM base	Direct
PDL1	PD-L1 Status	DM	DM base	Direct

F. STRATIFICATION VARIABLES

Variable	Label	Source (SDTM)	Merge Logic	Derivation Logic
STRAT1	PD-L1 Stratification	DM	DM base	High / Low

STRAT1N	PD-L1 (Numeric)	Stratification	DM	DM base	High=1, Low=2

G. ANALYSIS CONVENIENCE VARIABLES

Variable	Label	Source (SDTM)	Merge Logic	Derivation Logic
DTHFL	Death Flag	DS	ADSL + DS	Y if DSDECOD=DEATH
TRTGRP	Treatment Group	ADSL	Derived	Planned arm
STUDYDAY0	Baseline Reference Day	Derived	Derived	Day 0 reference

[Table 8.1: ADaM ADSL Dataset – Variable Derivation Specification (Complete Structure)]

◆ 8.4.2 ADTTE Derivation

The ADTTE dataset was derived to support time-to-event analysis such as Overall Survival (OS) and Progression-Free Survival (PFS). The ADTTE dataset was derived using ADSL and ADAE datasets.

PURPOSE

Used for survival analysis (OS, PFS, etc.).

Source Datasets

- ADSL – treatment start date
- ADVS (Analysis Data Vital Signs)

Key Variables Derived

- STARTDT = TRTSDT
- ADT = event date or censor date
- CNSR (0 = event, 1 = censored)
- ADY = ADT – STARTDT + 1

Merge Logic

ADSL (base – 1 record per subject)
LEFT JOIN

Event source (AE/DS)
BY USUBJID

ADVS → ADTTE
(Vitals-based Time-to-Event Analysis Design)

■ ADVS → ADTTE
Analysis Design & Derivation Specification

1. DATASET OVERVIEW

Item	Description
Dataset Name	ADTTE
Description	Time-to-Event Analysis Dataset derived from ADVS
Source Datasets	ADSL, ADVS
Purpose	To analyze time to first clinically significant vital sign event

2. POPULATION

Rule	Description
Analysis Population	Safety Population (SAFFL = "Y")
Treatment Start	TRTSDT (from ADSL)
Subjects Included	Subjects with ≥ 1 post-baseline vital assessment

3. EVENT DEFINITION

Event is defined as first post-baseline abnormal vital sign value per subject.

PARAMCD	Event Condition
SYSBP	AVAL ≥ 140
DIABP	AVAL ≥ 90
PULSE	AVAL < 50 OR AVAL > 120

Baseline records (ABLFL = "Y") are excluded. Only first occurrence is considered as an event.

4. TIME ORIGIN

Variable	Source	Definition
STARTDT	ADSL	TRTSDT (Treatment Start Date)

5. EVENT DATE

Variable	Source	Derivation
ADT	ADVS	Date of first qualifying post-baseline event

6. CENSORING RULES

Situation	CNSR	CNSDT
Event occurs	0	Event Date (ADT)
No event	1	Last available ADVS ADT
Early discontinuation	1	MIN(TRTEDT, Last ADVS ADT)

7. TIME-TO-EVENT CALCULATION

Variable	Derivation
AVAL	CNSDT – TRTSDT + 1
AVALU	"DAYS"

8. PARAMETER DERIVATION

Variable	Value
PARAMCD	Derived from ADVS.PARAMCD
PARAM	"Time to First Abnormal Vital Sign"

Rule	Description
Records per Subject	One record per subject per PARAMCD
Baseline Records	Excluded
Post-Baseline	Required
Missing Dates	Excluded from event determination

[Table 8.2: ADaM ADTTE Dataset – Time-to-Event Analysis Derivation Specification (OS & PFS)]

8.5 : DIGITAL APPENDIX - E : ADaM ANALYSIS DESIGN & DERIVATION

“Complete ADaM derivation specifications for all datasets are provided in the Digital Appendix (Appendix E).

▼ Content Available in Digital Appendix

- ✓ 01.ADSL (Analysis Data Subject Level)
- ✓ 02.ADAE (Analysis Data Adverse Events)
- ✓ 03.ADLB (Analysis Data Laboratory)
- ✓ 04.ADVS (Analysis Data Vital Signs)
- ✓ 05.ADTTE (Analysis Data Time-to-Event)
- ✓ 06.ADRESP (Analysis Data Response)

Digital Appendix Access Link : [GOOGLE DRIVE LINK]

https://drive.google.com/drive/folders/10wfur7DeqXXD_hqYmXuSovoCK0aX92ga?usp=drive_link



CHAPTER : 09

ADaM DATASET IMPLEMENTATION

◆ 9.0 INTRODUCTION

This chapter presents the implementation of ADaM datasets using SAS programming. The ADaM datasets were derived from SDTM datasets based on predefined analysis design and derivation specifications. Each dataset was generated using SAS code, validated through log verification, and reviewed using output datasets. This ensures accuracy, consistency, and readiness for statistical analysis.

◆ 9.1 ADAM DATASETS IMPLEMENTED

The following ADaM datasets were implemented in this study:

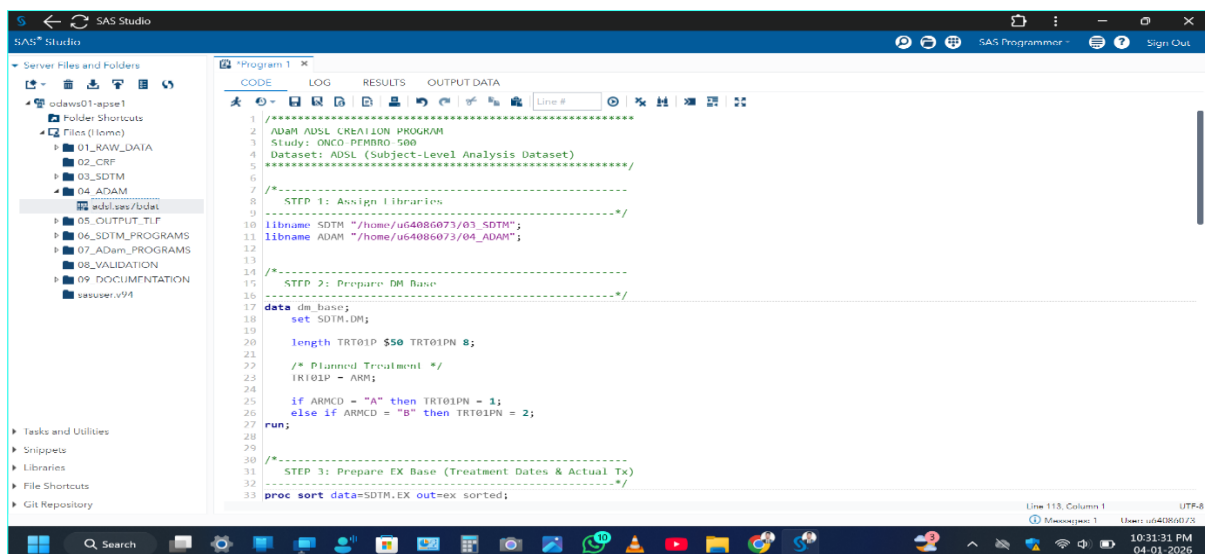
- ADSL – Subject-Level Analysis Dataset
- ADAE – Adverse Events Analysis Dataset
- ADLB – Laboratory Analysis Dataset
- ADVS – Vital Signs Analysis Dataset
- ADTTE – Time-to-Event Analysis Dataset
- ADRESP – Response Analysis Dataset

◆ 9.2 ADAM DATASET IMPLEMENTATION (DOMAIN-WISE DEMONSTRATION)

◆ 9.2.1 ADSL – Subject-Level Analysis Dataset

The ADSL dataset contains one record per subject and serves as the primary analysis dataset. It includes demographic details, treatment assignments, and analysis flags used across all downstream analyses.

👉 **Derived From: DM + DS + EX**



The screenshot displays the SAS Studio interface. On the left, the 'Server Files and Folders' pane shows a project structure with folders like '01_RAW_DATA', '02_CRF', '03_SDTM', '04_ADAM', '05_OUTPUT_TLF', '06_SDTM_PROGRAMS', '07_ADAM_PROGRAMS', '08_VALIDATION', '09_DOCUMENTATION', and 'sasuser.v94'. The '04_ADAM' folder is selected. The main window shows a SAS program with the following code:

```
1  /*****  
2  ADAM ADSL CREATION PROGRAM  
3  Study: ONCO PEMPRO 500  
4  Dataset: ADSL (Subject-Level Analysis Dataset)  
5  *****/  
6  
7  /*****  
8  STEP 1: Assign libraries  
9  *****/  
10 libname SDTM "/home/u64086073/03_SDTM";  
11 libname ADAM "/home/u64086073/04_ADAM";  
12  
13 /*****  
14 STEP 2: Prepare DM Base  
15 *****/  
16  
17 data dm_base;  
18   set SDTM.DM;  
19  
20   length TRT01P $50 TRT01PN 8;  
21  
22   /* Planned Treatment */  
23   TRT01P = ARM;  
24  
25   if ARMCD = "A" then TRT01PN = 1;  
26   else if ARMCD = "B" then TRT01PN = 2;  
27 run;  
28  
29 /*****  
30 STEP 3: Prepare EX Base (Treatment Dates & Actual Tx)  
31 *****/  
32  
33 proc sort data=SDTM.EX out=ex sorted;
```

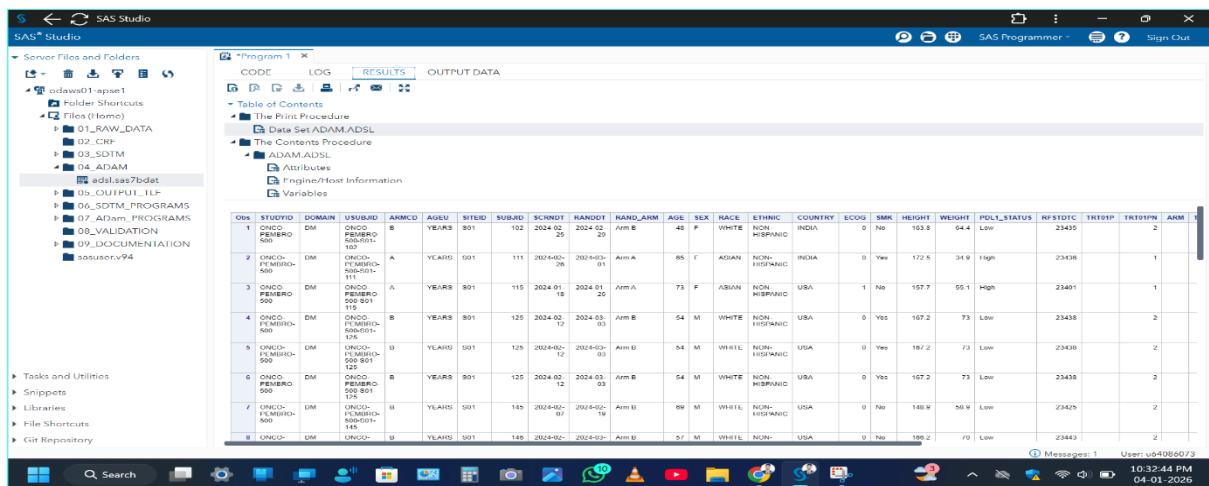
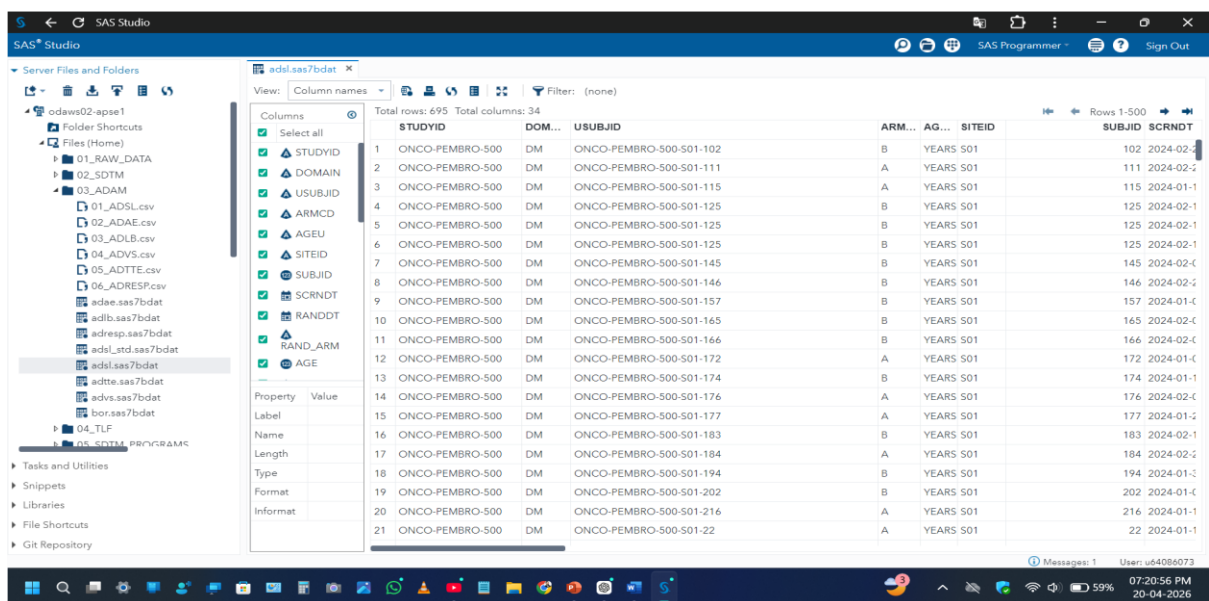


Figure 9.1: ADaM ADSL Dataset Implementation – SAS Code, Log, and Output Verification

9.2.2 Consolidated ADaM Dataset View

To verify the successful creation of all ADaM datasets, a consolidated SAS library view is presented below.

👉 [SAS LIBRARY SCREENSHOT SHOWING ALL 6 ADAM DATASETS]



[Figure 9.2: Consolidated SAS Library View – Verification of All ADaM Analysis Datasets]

[Complete ADaM datasets, including ADSL, ADAE, ADLB, ADVS, ADRESP, and ADTTE, along with corresponding SAS programs, execution logs, and output verification files, are provided in Digital Appendix – E for detailed reference and reproducibility.]

◆ 9.3 DIGITAL APPENDIX – F : ADaM DATASETS

▼ Content Available in Digital Appendix

The Digital Appendix includes comprehensive documentation for all ADaM datasets:

- ✓ All 06 ADaM datasets in SAS format (.sas7bdat)
- ✓ Corresponding CSV converted datasets
- ✓ Complete SAS programming code used for dataset generation
- ✓ Execution details including code, results, and logs
- ✓ Log files with zero errors and warnings
- ✓ SAS output dataset screenshots for verification

▼ Digital Appendix Access

👉 [INSERT GOOGLE DRIVE LINK]

https://drive.google.com/drive/folders/10r0K-WI-DVO_FyIGykTI6UEq0QSBQOuA?usp=drive_link



CHAPTER : 10

TLF SPECIFICATION & SHELL DESIGN

10.0 INTRODUCTION

In clinical trials, the final outputs used for interpretation, decision-making, and regulatory submission are Tables, Listings, and Figures (TLFs). These outputs are critical components of the Clinical Study Report (CSR) and provide structured representation of safety and efficacy results.

The TLF generation process follows a predefined workflow where outputs are first planned (blueprint), then structured (shell design), and finally generated using analysis datasets (ADaM).

In this study, a complete end-to-end TLF development process was implemented using SAS, ensuring alignment with CDISC standards and industry practices.

The objective of this chapter is to demonstrate how planned clinical outputs are systematically transformed into structured and reproducible TLFs using validated ADaM datasets.

10.1 TLF OUTPUT BLUEPRINT

Required Tables / Listings / Figures

TLF Blueprint defines **what outputs are required** before any programming begins.

This step ensures:

- No missing outputs
- Alignment with study objectives
- Regulatory readiness

10.1.1 Planned Outputs

- The TLF (Tables, Listings, and Figures) output blueprint defines all statistical outputs required for analysis and reporting. These outputs are generated using ADaM datasets and are aligned with clinical study reporting standards.
- The TLF blueprint was finalized prior to SAS programming to ensure consistency, traceability, and compliance with regulatory expectations.
- This chapter presents the design and specification of all Tables, Listings, and Figures (TLFs) developed for the study. These shells define the structure, variables, and statistical presentation format prior to actual programming and output generation.

Tables

1. Demographic and Baseline Characteristics
2. Treatment Exposure Summary
3. Treatment-Emergent Adverse Events Summary
4. Overall Response Rate (ORR)
5. Best Overall Response (BOR)

6. Laboratory Parameters Summary
7. Time-to-Event Endpoints

Listings

1. Subject Listing
2. Adverse Event Listing
3. Laboratory Results Listing
4. Tumor Response Listing
5. Time-to-Event Listing

Figures

1. Kaplan–Meier Plot (Overall Survival)
2. Kaplan–Meier Plot (Progression-Free Survival)
3. Bar Chart (Best Overall Response)

10.2 ADAM → TLF MAPPING : DATASET USAGE MAPPING :

TLFs were generated using validated ADaM datasets.

ADSL served as the core subject-level dataset and was consistently used across multiple TLF outputs for defining analysis populations.

10.2.1 Tables Mapping

Table No.	Table Name	ADaM Source
Table 1	Demographic & Baseline Characteristics	ADSL
Table 2	Treatment Exposure Summary	ADSL
Table 3	Treatment-Emergent Adverse Events Summary	ADSL + ADAE
Table 4	Overall Response Rate (ORR) Summary	ADSL + ADRESP
Table 5	Best Overall Response (BOR) Summary	ADSL + ADRESP
Table 6	Laboratory Parameters Summary	ADSL + ADLB
Table 7	Time-to-Event Endpoints (OS, PFS)	ADSL + ADTTE

[Table 10.1: ADaM to TLF Mapping for Tables (Dataset Usage Mapping)]

10.2.2 Listings Mapping

Listing No.	Listing Name	ADaM Source
Listing 1	Subject Listing	ADSL
Listing 2	Adverse Event Listing	ADSL + ADAE
Listing 3	Laboratory Results Listing	ADLB
Listing 4	Tumor Response Assessment Listing	ADSL + ADRESP
Listing 5	Time-to-Event Listing	ADSL + ADTTE

[Table 10.2: ADaM to TLF Mapping for Listings (Dataset Usage Mapping)]

10.2.3 Figures Mapping

Figure No.	Figure Name	ADaM Source
Figure 1	Kaplan–Meier Plot of Overall Survival (OS)	ADTTE
Figure 2	Kaplan–Meier Plot of Progression-Free Survival (PFS)	ADTTE
Figure 3	Bar Chart of Best Overall Response (BOR)	ADRESP

[Table 10.3: ADaM to TLF Mapping for Figures (Dataset Usage Mapping)]

CHAPTER : 11

STATISTICAL ANALYSIS & TLF GENERATION

◆ 11.0 INTRODUCTION

This chapter presents the statistical analysis performed and the generation of Tables, Listings, and Figures (TLFs) using SAS programming. The outputs were created using validated ADaM datasets and were designed according to predefined TLF specifications.

The objective of this chapter is to demonstrate the practical implementation of statistical programming workflows, including code execution, output generation, and validation.

All statistical outputs were generated using SAS OnDemand environment and exported in Rich Text Format (RTF) using ODS for reporting purposes

◆ 11.1 STATISTICAL ANALYSIS OVERVIEW

The statistical analysis performed in this study includes:

◆ Descriptive Analysis

- Demographic and baseline characteristics
- Laboratory parameters
- Treatment exposure

◆ Safety Analysis

- Treatment-Emergent Adverse Events (TEAE)
- Severity and seriousness of adverse events

◆ Efficacy Analysis

- Overall Response Rate (ORR)
- Best Overall Response (BOR)
- Time-to-Event Endpoints (OS, PFS)

◆ Survival Analysis

- Kaplan–Meier method
- Log-rank test for comparison between treatment arms

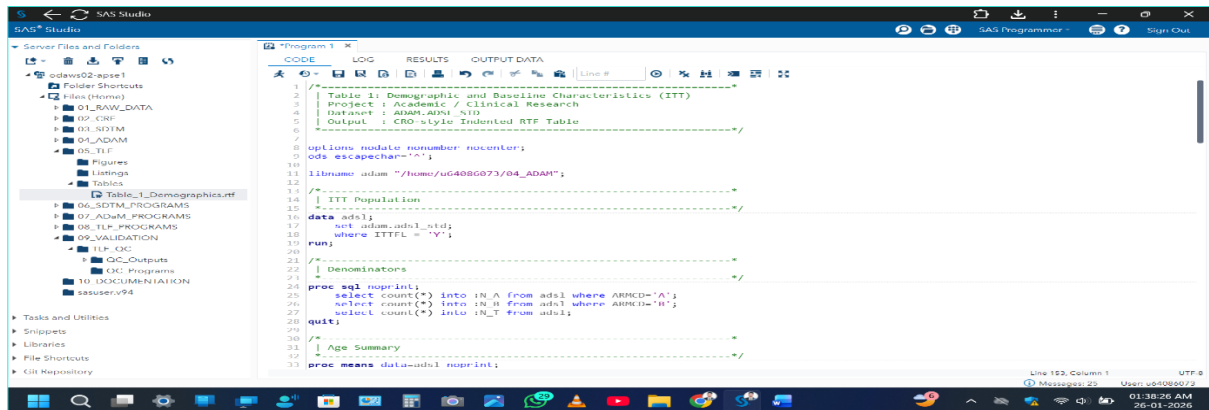
All analyses were performed using standard SAS procedures such as PROC MEANS, PROC FREQ, PROC LIFETEST, and PROC SGPLOT.

11.2 TLF GENERATION USING SAS

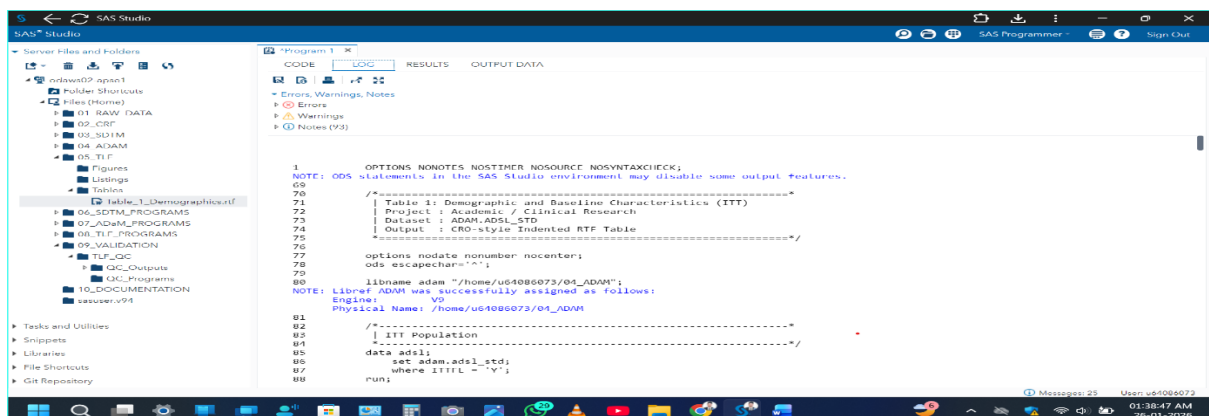
◆ 11.2.1 TABLES (SUMMARY OUTPUTS)

Table 1: Demographic and Baseline Characteristics

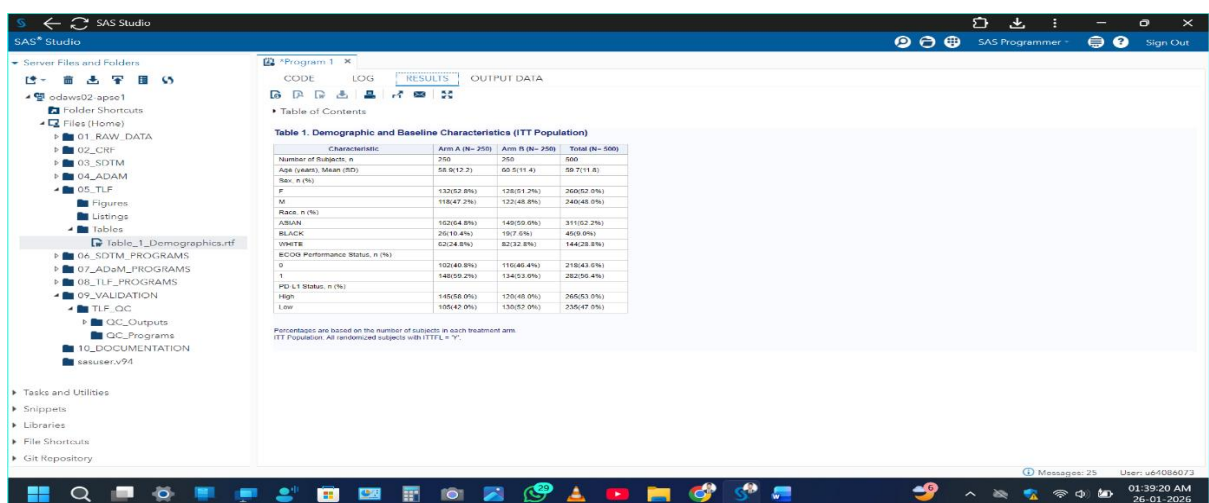
- To summarize baseline demographic characteristics of subjects across treatment arms.



```
1  /*-----*/
2  | Table 1: Demographic and Baseline Characteristics (ITT)
3  | Project : Academic / Clinical Research
4  | Dataset : ADAM.ADSL_STD
5  | Output  : CRO-style Indented RTF Table
6  |-----*/
7
8  options nodate nonumber nocenter;
9  ods escapechar='^';
10
11  libname adam "/home/u64086073/04_ADAM";
12
13  /*-----*/
14  | ITT Population
15  |-----*/
16  data adsl;
17    set adam.adsl_std;
18    where ITTFL = 'Y';
19  run;
20
21  /*-----*/
22  | Denominators
23  |-----*/
24
25  proc sql noprint;
26    select count(*) into :N_A from adsl where ARRCO='A';
27    select count(*) into :N_B from adsl where ARRCO='B';
28    select count(*) into :N_T from adsl;
29  quit;
30
31  /*-----*/
32  | Age Summary
33  |-----*/
34  proc means data=adsl noprint;
```



```
1  NOTE: ODS statements in the SAS Studio environment may disable some output features.
69
70  /*-----*/
71  | Table 1: Demographic and Baseline Characteristics (ITT)
72  | Project : Academic / Clinical Research
73  | Dataset : ADAM.ADSL_STD
74  | Output  : CRO-style Indented RTF Table
75  |-----*/
76
77  options nodate nonumber nocenter;
78  ods escapechar='^';
79
80  libname adam "/home/u64086073/04_ADAM";
81  NOTE: Libref ADAM was successfully assigned as follows:
82  Engine: V9
83  Physical Name: /home/u64086073/04_ADAM
84
85  /*-----*/
86  | ITT Population
87  |-----*/
88  data adsl;
89    set adam.adsl_std;
90    where ITTFL = 'Y';
91  run;
```



Characteristic	Arm A (N= 250)	Arm B (N= 250)	Total (N= 500)
Number of Subjects: n	250	250	500
Age (years), Mean (SD)	55.9(12.7)	60.5(11.4)	59.7(11.8)
Sex: n (%)			
F	122(52.8%)	128(51.2%)	250(52.0%)
M	118(47.2%)	122(48.8%)	240(48.0%)
Race: n (%)			
ASIAN	102(40.8%)	140(56.0%)	242(48.2%)
BLACK	26(10.4%)	19(7.6%)	45(9.0%)
WHITE	63(24.8%)	82(32.8%)	144(28.8%)
ECOG Performance Status, n (%)			
0	102(40.8%)	116(46.4%)	218(43.6%)
1	148(59.2%)	134(53.6%)	282(56.4%)
HQ L1 Status, n (%)			
High	145(58.0%)	150(60.0%)	295(59.0%)
Low	105(42.0%)	100(40.0%)	205(41.0%)

Percentages are based on the number of subjects in each treatment arm.
ITT Population: All randomized subjects with ITTFL = 'Y'.

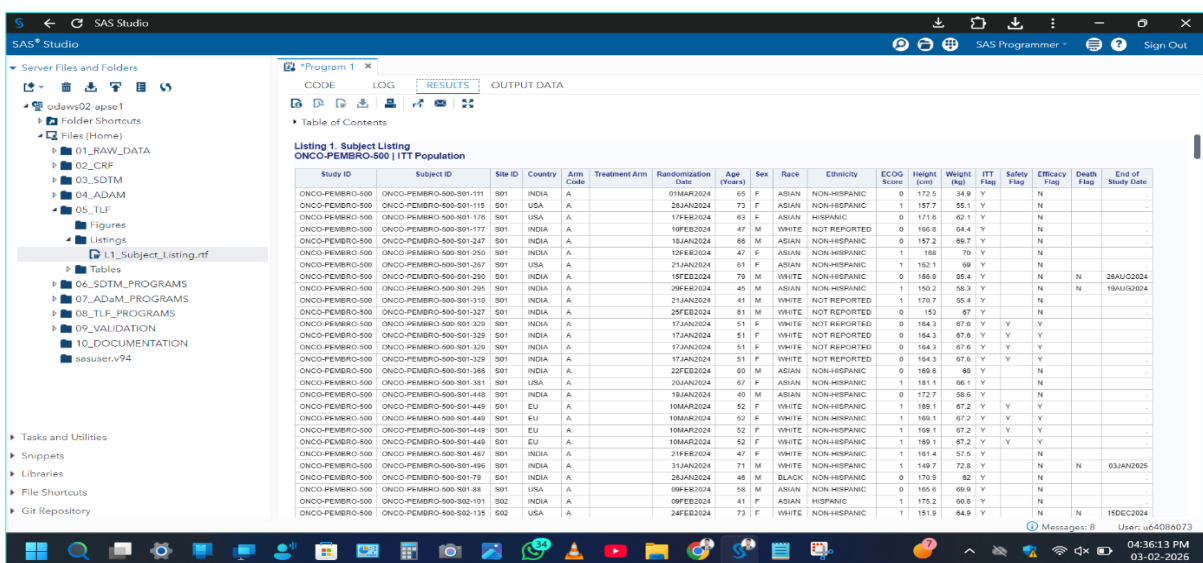
[Table 11.1: Demographic and Baseline Characteristics]

◆ 11.2.2 LISTINGS (SUBJECT-LEVEL DATA)

Listing 1: Subject Listing

Description:

This table presents summary statistics including median survival time, minimum, and maximum values.



Study ID	Subject ID	Site ID	Country	Arm Code	Treatment Arm	Randomization Date	Age (Years)	Sex	Race	Ethnicity	ECOG Score	Height (cm)	Weight (kg)	ITT Flag	Safety Flag	Efficacy Flag	Death Flag	End of Study Date
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-111	501	INDIA	A		01MAR2024	60	F	ASIAN	NON-HISPANIC	0	172.5	54.9	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-115	501	USA	A		25JAN2024	73	F	ASIAN	NON-HISPANIC	1	157.7	55.1	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-176	501	USA	A		17SEP2024	63	F	ASIAN	HISPANIC	0	171.6	62.1	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-177	501	INDIA	A		19FEB2024	47	M	WHITE	NOT REPORTED	0	165.9	64.4	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-247	501	INDIA	A		18JAN2024	66	M	ASIAN	NON-HISPANIC	0	157.2	56.7	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-250	501	INDIA	A		12SEP2024	47	F	ASIAN	NON-HISPANIC	1	168	70	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-267	501	USA	A		21JAN2024	61	F	ASIAN	NON-HISPANIC	1	162.1	69	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-290	501	INDIA	A		19FEB2024	79	M	WHITE	NON-HISPANIC	0	166.9	65.4	Y	N		N	28AUG2024
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-298	501	INDIA	A		29APR2024	45	M	ASIAN	NON-HISPANIC	1	160.2	58.3	Y	N		N	19AUG2024
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-310	501	INDIA	A		21JAN2024	41	M	WHITE	NOT REPORTED	1	170.7	55.4	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-327	501	INDIA	A		25FEB2024	61	M	WHITE	NOT REPORTED	0	153	67	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-329	501	INDIA	A		17JAN2024	51	F	WHITE	NOT REPORTED	0	164.3	67.6	Y	Y	Y		
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-329	501	INDIA	A		17JAN2024	51	F	WHITE	NOT REPORTED	0	164.3	67.6	Y	Y	Y		
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-329	501	INDIA	A		17JAN2024	51	F	WHITE	NOT REPORTED	0	164.3	67.6	Y	Y	Y		
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-329	501	INDIA	A		17JAN2024	51	F	WHITE	NOT REPORTED	0	164.3	67.6	Y	Y	Y		
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-386	501	INDIA	A		22FEB2024	60	M	ASIAN	NON-HISPANIC	0	166.6	66	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-381	501	USA	A		20JAN2024	67	F	ASIAN	NON-HISPANIC	1	161.1	56.1	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-448	501	INDIA	A		19JAN2024	40	M	ASIAN	NON-HISPANIC	0	172.7	58.6	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-449	501	EU	A		10MAR2024	52	F	WHITE	NON-HISPANIC	1	169.1	67.2	Y	Y	Y		
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-449	501	EU	A		10MAR2024	52	F	WHITE	NON-HISPANIC	1	169.1	67.2	Y	Y	Y		
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-449	501	EU	A		10MAR2024	52	F	WHITE	NON-HISPANIC	1	169.1	67.2	Y	Y	Y		
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-457	501	INDIA	A		21FEB2024	47	F	WHITE	NON-HISPANIC	1	161.4	57.5	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-466	501	INDIA	A		31JAN2024	71	M	WHITE	NON-HISPANIC	1	149.7	72.8	Y	N		N	03JAN2025
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-79	501	INDIA	A		25JAN2024	46	M	BLACK	NON-HISPANIC	0	170.5	82	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-501-88	501	USA	A		09SEP2024	58	M	ASIAN	NON-HISPANIC	0	165.6	69.9	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-502-101	502	INDIA	A		09FEB2024	41	F	ASIAN	HISPANIC	1	175.2	60.8	Y	N			
ONCO-PEMBRO-500	ONCO-PEMBRO-500-502-135	502	USA	A		24FEB2024	73	F	WHITE	NON-HISPANIC	1	151.9	64.9	Y	N		N	15DEC2024

[Listing 11.1: Subject-Level Listing]

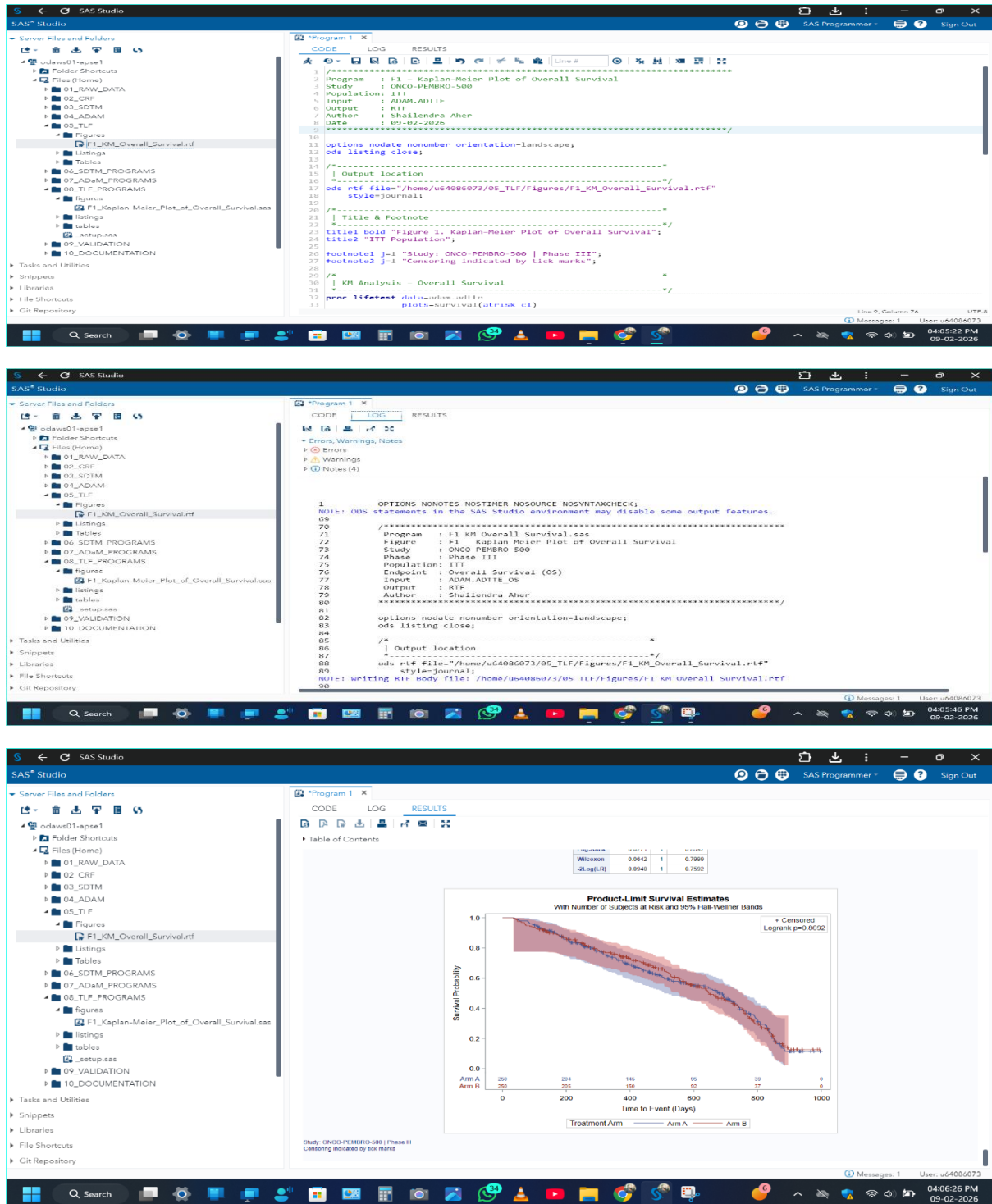
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◆ 11.2.3 FIGURES (GRAPHICAL OUTPUTS)

Figure 1: Kaplan–Meier Plot (OS)

Purpose:

To visualize overall survival probability over time.

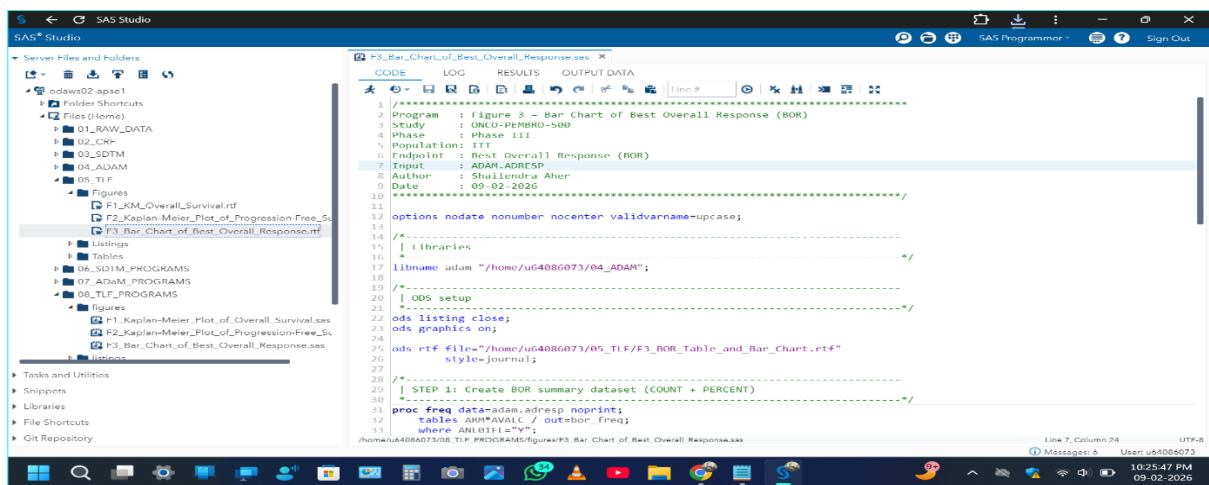


[Figure 11.1: Kaplan–Meier Plot of Overall Survival (OS)]

Figure 3: BOR Bar Chart

Purpose:

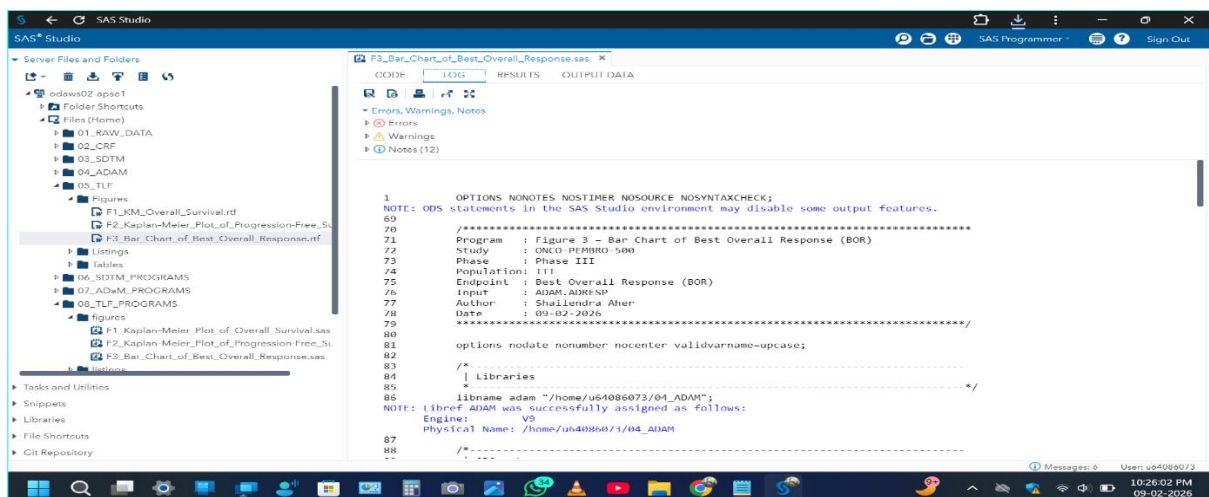
To visually compare response distribution across treatment arms.



```

1  /*****
2  Program : Figure 3 - Bar Chart of Best Overall Response (BOR)
3  Study : ONCO-PEMBRO-500
4  Phase : Phase III
5  Population: ITT
6  Endpoint : Best Overall Response (BOR)
7  Input : ADAM.ADRRESP
8  Author : Shailendra Aher
9  Date : 09-02-2026
10 *****/
11
12 options nodate nonumber nocenter validvarname=upcase;
13
14 /*****
15 Libraries
16 *****/
17 libname adam "/home/u64086073/04_ADAM";
18
19 /*****
20 ODS setup
21 *****/
22 ods listing close;
23 ods graphics on;
24
25 ods rtf file="/home/u64086073/05_TLF/F3_BOR_Table_and_Bar_Chart.rtf"
26 style=journal;
27
28 /*****
29 STEP 1: Create BOR summary dataset (COUNT + PERCENT)
30 *****/
31 proc freq data=adam.adrresp noprint;
32 tables ARM*AVAILC / out=bor_freq;
33 where ANL01F1="V";
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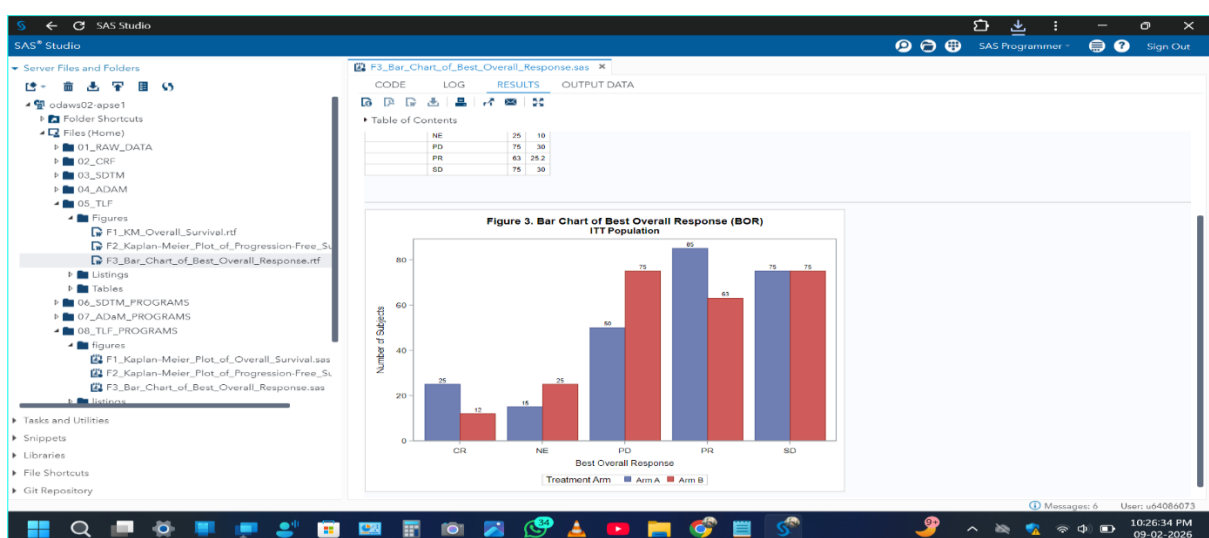
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```

1  OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
2  NOTE: ODS statements in the SAS Studio environment may disable some output features.
3
4  /*****
5  Program : Figure 3 - Bar Chart of Best Overall Response (BOR)
6  Study : ONCO-PEMBRO-500
7  Phase : Phase III
8  Population: ITT
9  Endpoint : Best Overall Response (BOR)
10 Input : ADAM.ADRRESP
11 Author : Shailendra Aher
12 Date : 09-02-2026
13 *****/
14
15 options nodate nonumber nocenter validvarname=upcase;
16
17 /*****
18 Libraries
19 *****/
20 libname adam "/home/u64086073/04_ADAM";
21 NOTE: libref ADAM was successfully assigned as follows:
22 Engines: V9
23 Physical Name: /home/u64086073/04_ADAM
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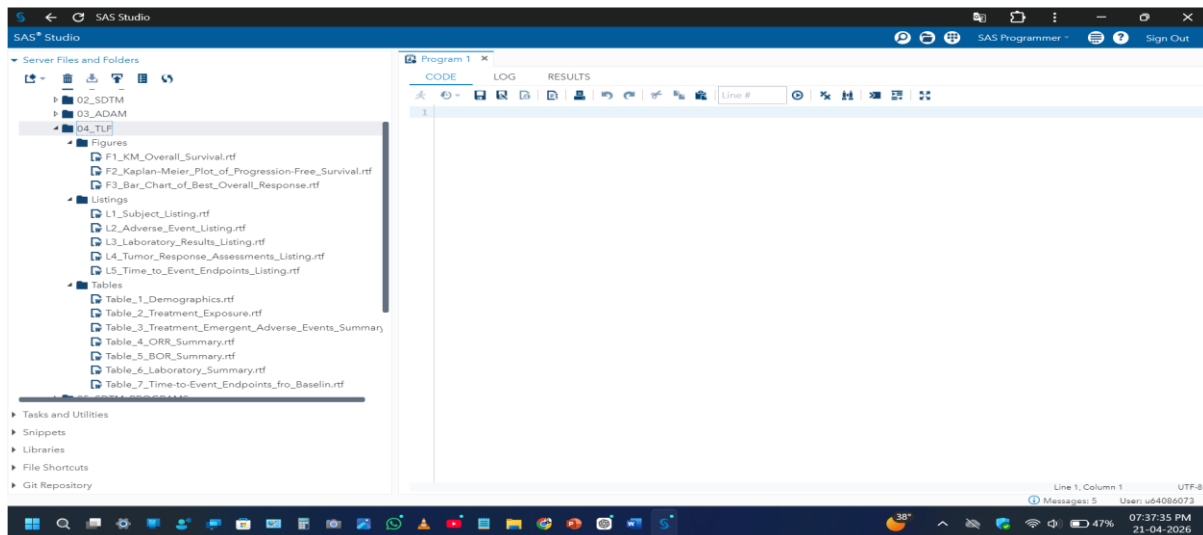
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[FIGURE 11.2: BAR CHART OF BEST OVERALL RESPONSE (BOR)]

◆ 11.3 CONSOLIDATED OUTPUT VIEW

The following screenshot shows all 15 generated TLF outputs within the SAS environment, confirming successful execution of the complete statistical workflow. To verify successful generation of all outputs, a consolidated SAS output view is presented.



[Figure 11.4: Consolidated SAS Output View Showing All Generated TLF Outputs]

❖ 11.4 TRACEABILITY & DATA FLOW

- Traceability is a critical requirement in clinical data analysis, ensuring that every output can be linked back to its source datasets. In this study, full traceability was maintained across the data pipeline : Raw Data → SDTM → ADaM → TLF Outputs
- Each TLF output was generated using validated ADaM datasets, which were derived from standardized SDTM domains. The mapping between ADaM datasets and TLF outputs ensures that all results are reproducible and verifiable.

Data Stage	Total Files	Key Components / Domains	Industry Standard
1. Raw Input	09 Files	DM, AE, EX, LB, VS, TU, TR, RS, DS	CDM Export (CSV)
2. SDTM	14 Domains	DM, AE, EX, LB, VS, TU, TR, RS, DS + 5 SUPP Domains	CDISC SDTM v3.2
3. ADaM	06 Datasets	ADSL, ADAE, ADLB, ADVS, ADRESP, ADTTE	CDISC ADaM v1.1
4. TLF Output	15 Outputs	07 Tables, 05 Listings, 03 Figures (KM Plot)	ICH E3 Guidelines

[TABLE 11.8: END-TO-END DATA TRACEABILITY AND PIPELINE FLOW]

◆ 11.5 : DIGITAL APPENDIX – G : TLF STATISTICAL OUTPUT

This ensures full transparency, reproducibility, and audit readiness of all statistical outputs

Due to the large volume of outputs and supporting files, complete documentation is provided in the Digital Appendix.G

▼ Content Available in Digital Appendix

The Digital Appendix includes:

- ✓ All 15 TLF outputs in RTF format
- ✓ Complete SAS programming code
- ✓ Execution logs (zero errors and warnings)
- ✓ Execution details including code, results, and logs

▼ Digital Appendix Access

👉 [GOOGLE DRIVE LINK]

https://drive.google.com/drive/folders/1ObwVPTpggETL54DOyxd8AVHMtrI-fBdC?usp=drive_link



CHAPTER : 12
QUALITY CONTROL
& VALIDATION

◆ 12.1 : INTRODUCTION

Quality Control (QC) and validation are essential components of clinical data analysis to ensure that all statistical outputs are accurate, reliable, and reproducible. In clinical research, even minor errors in statistical programming can lead to incorrect interpretation of results, which may impact regulatory decisions.

Therefore, a robust validation framework is required to confirm that all analyses are performed correctly and consistently.

In this study, QC was implemented using a structured and systematic approach aligned with industry practices. The validation process focused on ensuring:

- Accuracy of statistical results
- Consistency across outputs
- Reproducibility of analysis
- Compliance with clinical data standards

The Independent Double Programming (IDP) method was used as the primary validation approach. This method is widely adopted in the pharmaceutical industry as it provides a strong mechanism to detect programming errors and validate outputs independently.

The objective of QC in this study is to ensure that all generated outputs are correct, traceable, and free from programming discrepancies.

◆ 12.2 : VALIDATION METHODOLOGY: INDEPENDENT DOUBLE PROGRAMMING (IDP)

Independent Double Programming (IDP) is a standard validation technique used in clinical statistical programming to ensure the accuracy of analysis outputs.

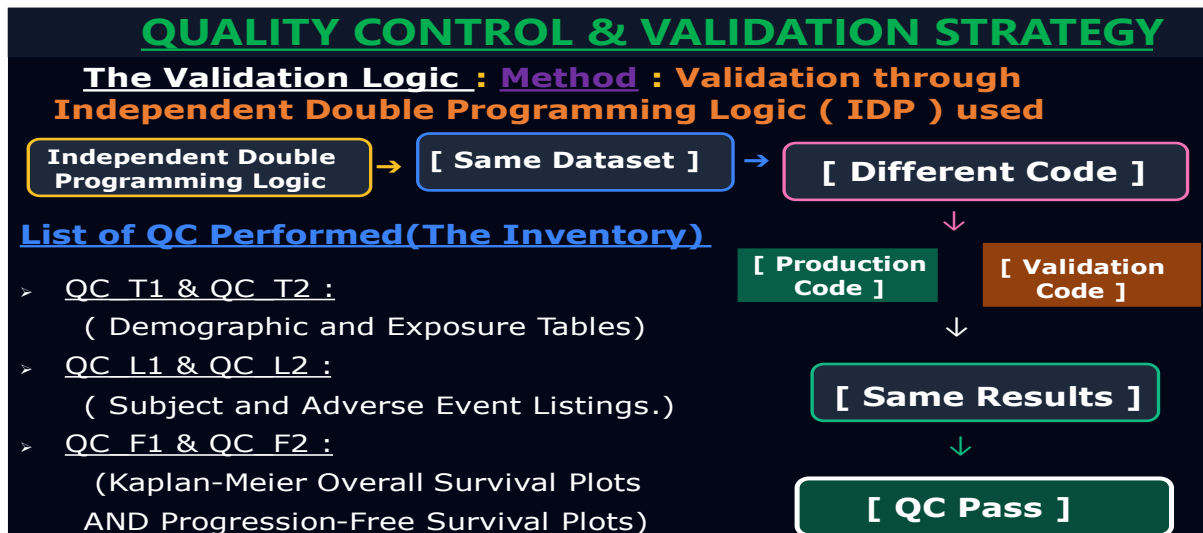
In this approach, two separate programs are developed independently using the same input datasets:

- The first program is referred to as the **Production Program**, which generates the primary outputs.
- The second program is referred to as the **Validation or QC Program**, which is developed independently using different logic or coding style.

The key principle of IDP is that both programs must produce identical results despite being independently developed.

The process involves:

1. Using the same input dataset (ADaM datasets)
2. Writing independent code for production and validation
3. Generating outputs separately
4. Comparing results between the two outputs



[Figure 12.1: Quality Control and Validation Strategy Using Independent Double Programming (IDP) Method]

◆ 12.3 QC SCOPE AND VALIDATION STRATEGY (EXPANDED)

Validation of these representative outputs ensures confidence in the overall programming logic and consistency across all TLF outputs.

In large-scale clinical studies, performing QC for all outputs may not always be feasible due to time and resource constraints. Therefore, a selective validation strategy is often applied.

In this study, QC was performed on a representative subset of TLF outputs rather than all 15 outputs.

The selected outputs include:

◆ Tables

- Table 1: Demographic and Baseline Characteristics
- Table 2: Treatment Exposure

◆ Listings

- Listing 1: Subject Listing
- Listing 2: Adverse Event Listing

◆ Figures

- Figure 1: Kaplan–Meier Plot of Overall Survival
- Figure 2: Kaplan–Meier Plot of Progression-Free Survival

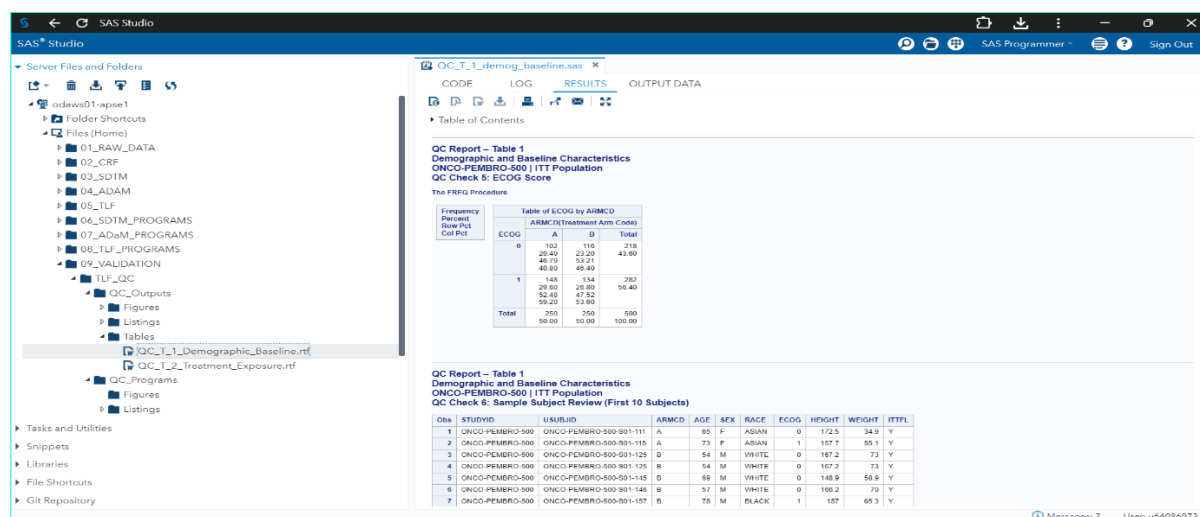
These outputs were selected because they represent different domains of analysis:

- Baseline characteristics , Treatment exposure , Safety evaluation , Survival analysis

12.4 QC RESULTS & VALIDATION PROOF

◆ QC for Table 1: Demographics and baseline

QC Comparison: The output generated using independent validation programming was compared with the production output. All values including subject counts, percentages, and summary statistics matched exactly. No discrepancies were observed, confirming QC PASS.



[Figure 12.2: QC Validation Comparison for Table 1 – Demographic and Baseline Characteristics (Production vs Validation Outputs)]

👉 GROUP JUSTIFICATION : QC Scope Justification (Tables):

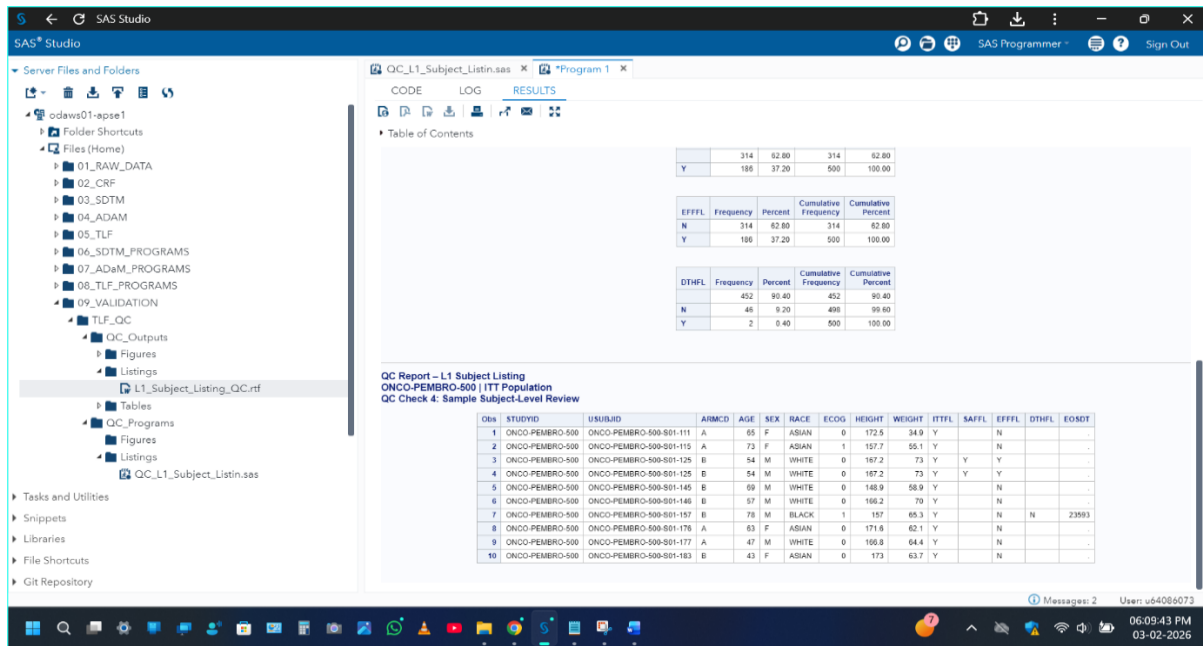
➤ QC was performed **for Table 1 and Table 2** as they represent key baseline and exposure analyses. Since these tables form the foundation for subject-level and treatment-related summaries, successful validation ensures confidence in similar table structures. Therefore, **The remaining tables:**

- Table 3 (Adverse Events Summary)
- Table 4 (Overall Response Rate)
- Table 5 (Best Overall Response)
- Table 6 (Laboratory Summary)
- Table 7 (Time-to-Event Endpoints)

- follow similar derivation logic and structural patterns based on validated ADaM datasets (ADAE, ADRESP, ADLB, ADTTE).
- Therefore, successful QC validation of Table 1 and Table 2 provides sufficient confidence in the correctness of programming logic applied across other summary tables.

◆ QC for Listing 1: Subject Listing

QC Comparison: All subject-level records including demographic and treatment details matched exactly between production and validation outputs, confirming QC PASS.



[Figure 12.4: QC Validation Comparison for Listing 1 – Subject Listing (Production vs Validation Outputs)]

👉 GROUP JUSTIFICATION (Listings)

QC Scope Justification (Listings):

- QC was performed for Listing 1 and Listing 2 as they represent subject-level and safety listings. These listings cover core structural logic applicable to other listings. Therefore, additional listing QC was not performed within the project scope.

These listings demonstrate:

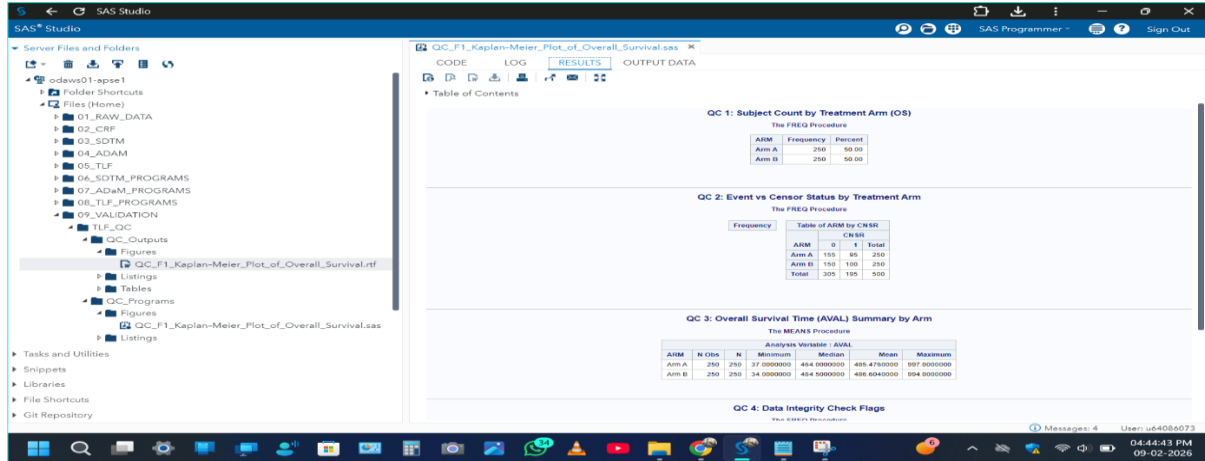
- Data structure handling
- Variable mapping
- Subject-level traceability

The remaining listings:

- Listing 3 (Laboratory Results Listing)
- Listing 4 (Tumor Response Listing)
- Listing 5 (Time-to-Event Listing)

◆ QC for Figure 1: Overall Survival (KM Plot)

QC Comparison: The Kaplan–Meier survival curves and associated statistics generated using validation programming were consistent with production outputs. No differences were observed, confirming QC PASS.



[Figure 12.6: QC Validation Comparison for Figure 1 – Overall Survival (Kaplan–Meier Plot)]

QC Scope Justification (Figures – FINAL ENHANCED)

QC was performed for:

- Figure 1 (Overall Survival – Kaplan–Meier Plot)
- Figure 2 (Progression-Free Survival – Kaplan–Meier Plot)

These figures represent key survival endpoints and involve statistical procedures such as Kaplan–Meier estimation and log-rank testing.

❖ The remaining figure:

- Figure 3 (Best Overall Response Bar Chart)
 - is generated using standard graphical procedures (PROC SGPLOT) and is based on validated ADRESP dataset.
 - Since similar graphical programming logic is used, validation of key survival figures ensures correctness of other graphical outputs
- ✓ Overall, QC was strategically performed on representative outputs covering baseline, safety, and efficacy domains. This ensures validation of core programming logic across all TLF outputs, while maintaining practical feasibility within academic **project constraints**.

12.5 : FINAL QC SUMMARY & VALIDATION OUTCOME

◆ Overview

This section presents a consolidated summary of the Quality Control (QC) validation performed for key statistical outputs generated in this study.

The QC process was executed using the Independent Double Programming (IDP) approach, where outputs generated from production programs were independently reproduced using separate validation programs. The results from both programs were systematically compared to ensure consistency, accuracy, and reproducibility.

The comparison focused on verifying:

- Subject counts and percentages
- Summary statistics (Mean, Median, Min, Max)
- Subject-level listing records
- Survival analysis outputs and graphical representations

All comparisons were performed using a combination of statistical procedures (PROC FREQ, PROC SQL, PROC MEANS) and visual verification techniques.

◆ QC Summary Table

Output	Comparison Result	QC Status
Table 1 (Demographics)	All values matched	PASS
Table 2 (Treatment Exposure)	All values matched	PASS
Listing 1 (Subject Listing)	All records matched	PASS
Listing 2 (Adverse Event Listing)	All records matched	PASS
Figure 1 (Overall Survival)	Curves and estimates matched	PASS
Figure 2 (Progression-Free Survival)	Curves and estimates matched	PASS

[Table 12.1: Final QC Summary and Validation Outcome of All TLF Outputs (Independent Double Programming Results)]

◆ Interpretation of QC Results

The above summary confirms that all validated outputs demonstrated complete agreement between production and validation programs.

◆ 12.6 : APPENDIX - H : QUALITY CONTROL & VALIDATION

- Due to the large volume of QC documentation, detailed validation materials are provided in the Digital Appendix.
- This ensures full transparency, audit readiness, and reproducibility of the validation process

❖ Content Available in Digital Appendix H

- ✓ QC validation screenshots for selected outputs (T1, T2, L1, L2, F1, F2)
- ✓ Complete SAS programming code for both production and QC programs
- ✓ Execution logs confirming zero errors and warnings
- ✓ Detailed execution records including code, results, and logs

▼ Digital Appendix Access

👉 [GOOGLE DRIVE LINK]

https://drive.google.com/drive/folders/1wpytpQAqrHV-RylSz_yUB5tzyIHIFZwu?usp=drive_link



◆ 12.7 : SUMMARY

- This chapter presented the implementation of quality control and validation using the Independent Double Programming method.
 - The validation process confirmed that the statistical outputs generated in this study are accurate, consistent, and reproducible.
 - The successful validation of representative outputs demonstrates the robustness of the programming workflow and ensures confidence in the overall analysis
-

CHAPTER : 13

RESULTS & DISCUSSION

◆ 13.1 INTRODUCTION

This chapter presents the statistical results generated from the synthetic clinical trial data using the SAS-based CDISC-compliant workflow.

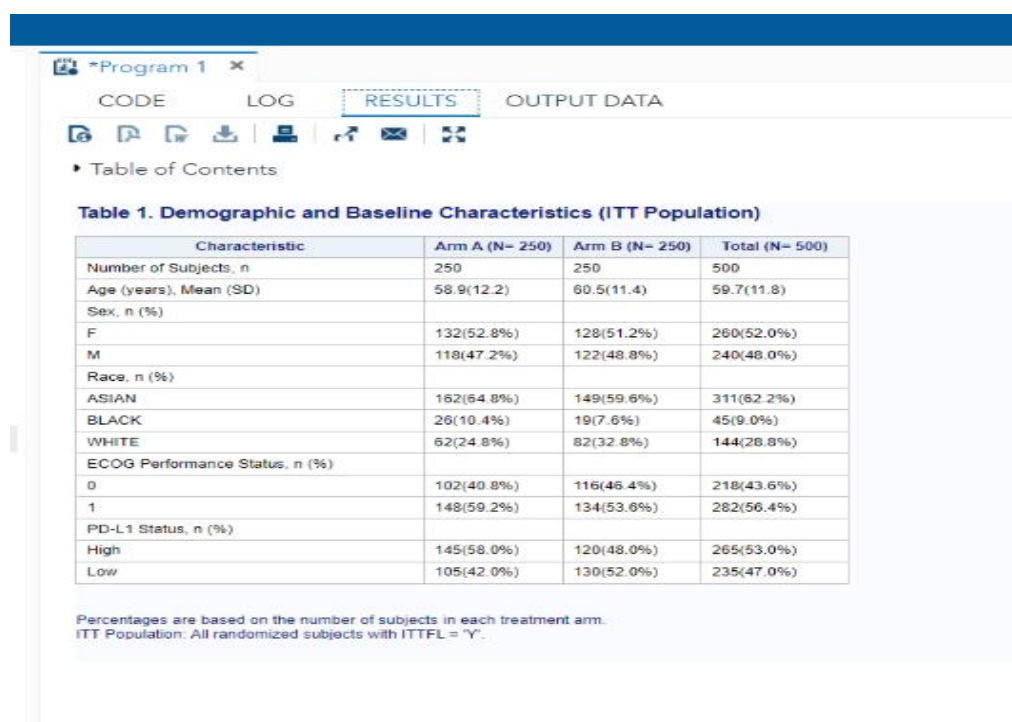
The objective of this chapter is to demonstrate the successful transformation of data from raw datasets to final Tables, Listings, and Figures (TLFs), and to interpret the outputs generated through statistical programming.

The focus is on:

- Accuracy of results
- Consistency of outputs
- Successful implementation of the SAS pipeline

This chapter emphasizes the validation of the data processing workflow rather than clinical efficacy conclusions.

◆ 13.2 DEMOGRAPHIC & BASELINE CHARACTERISTICS



The screenshot shows the SAS Studio interface with the 'RESULTS' tab selected. A table titled 'Table 1. Demographic and Baseline Characteristics (ITT Population)' is displayed. The table has four columns: 'Characteristic', 'Arm A (N= 250)', 'Arm B (N= 250)', and 'Total (N= 500)'. The rows list various demographic and baseline characteristics with their respective counts and percentages for each arm and the total population. Below the table, a note states: 'Percentages are based on the number of subjects in each treatment arm. ITT Population: All randomized subjects with ITTFL = "Y".'

Characteristic	Arm A (N= 250)	Arm B (N= 250)	Total (N= 500)
Number of Subjects, n	250	250	500
Age (years), Mean (SD)	58.9(12.2)	60.5(11.4)	59.7(11.8)
Sex, n (%)			
F	132(52.8%)	128(51.2%)	260(52.0%)
M	118(47.2%)	122(48.8%)	240(48.0%)
Race, n (%)			
ASIAN	162(64.8%)	149(59.6%)	311(62.2%)
BLACK	26(10.4%)	19(7.6%)	45(9.0%)
WHITE	62(24.8%)	82(32.8%)	144(28.8%)
ECOG Performance Status, n (%)			
0	102(40.8%)	116(46.4%)	218(43.6%)
1	148(59.2%)	134(53.6%)	282(56.4%)
PD-L1 Status, n (%)			
High	145(58.0%)	120(48.0%)	265(53.0%)
Low	105(42.0%)	130(52.0%)	235(47.0%)

[Table 13.1: Demographic and Baseline Characteristics (Final TLF Output – ITT Population)]

◆ Results Summary

A total of 500 subjects were included in the study, with 250 subjects in each treatment arm.

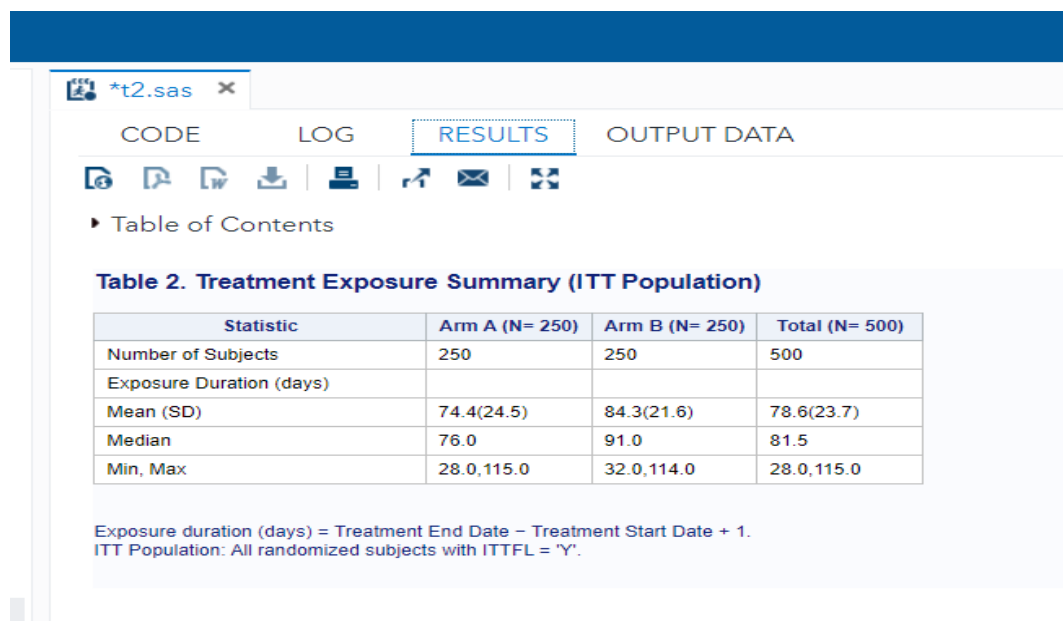
The demographic characteristics were well balanced between the two groups:

- Mean age was comparable across both arms
- Gender distribution was approximately equal
- Majority of subjects belonged to the Asian population
- ECOG performance status showed similar distribution
- PD-L1 status was balanced across treatment groups

◆ Interpretation

- The similarity in baseline characteristics between treatment arms confirms that the randomization process was correctly implemented in the synthetic dataset.
- This ensures that any downstream analysis is not biased due to population imbalance.
- This demonstrates accurate subject-level data handling and correct implementation of demographic analysis programming.

◆ 13.3 TREATMENT EXPOSURE ANALYSIS (SAFETY BASE)



Statistic	Arm A (N= 250)	Arm B (N= 250)	Total (N= 500)
Number of Subjects	250	250	500
Exposure Duration (days)			
Mean (SD)	74.4(24.5)	84.3(21.6)	78.6(23.7)
Median	76.0	91.0	81.5
Min, Max	28.0,115.0	32.0,114.0	28.0,115.0

Exposure duration (days) = Treatment End Date - Treatment Start Date + 1.
ITT Population: All randomized subjects with ITTFL = 'Y'.

[Table 13.2: Treatment Exposure Summary (Final TLF Output – ITT Population)]

◆ Results Summary

- Mean exposure duration was slightly higher in one treatment arm
- Median exposure values were consistent

- Minimum and maximum exposure durations were within expected ranges

◆ Interpretation

- The exposure analysis confirms that treatment administration data has been correctly derived and summarized.
- This validates the correctness of date handling, duration calculation, and statistical summarization logic.
- Accurate exposure calculations indicate correct transformation from SDTM EX domain to ADaM datasets.

◆ 13.4 : SAFETY ANALYSIS (ADVERSE EVENTS)

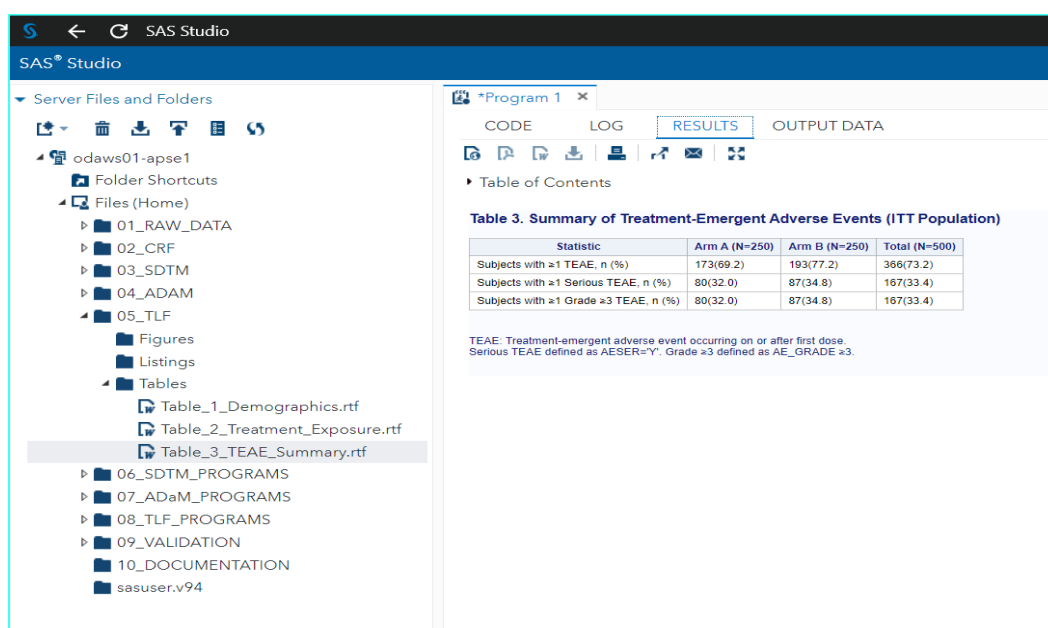


Table 3. Summary of Treatment-Emergent Adverse Events (ITT Population)

Statistic	Arm A (N=250)	Arm B (N=250)	Total (N=500)
Subjects with ≥1 TEAE, n (%)	173(69.2)	193(77.2)	366(73.2)
Subjects with ≥1 Serious TEAE, n (%)	80(32.0)	87(34.8)	167(33.4)
Subjects with ≥1 Grade ≥3 TEAE, n (%)	80(32.0)	87(34.8)	167(33.4)

TEAE: Treatment-emergent adverse event occurring on or after first dose.
Serious TEAE defined as AESER="Y". Grade ≥3 defined as AE_GRADE ≥3.

[Table 13.3: Summary of Treatment-Emergent Adverse Events (Final TLF Output – Safety Analysis)]

◆ Results Summary

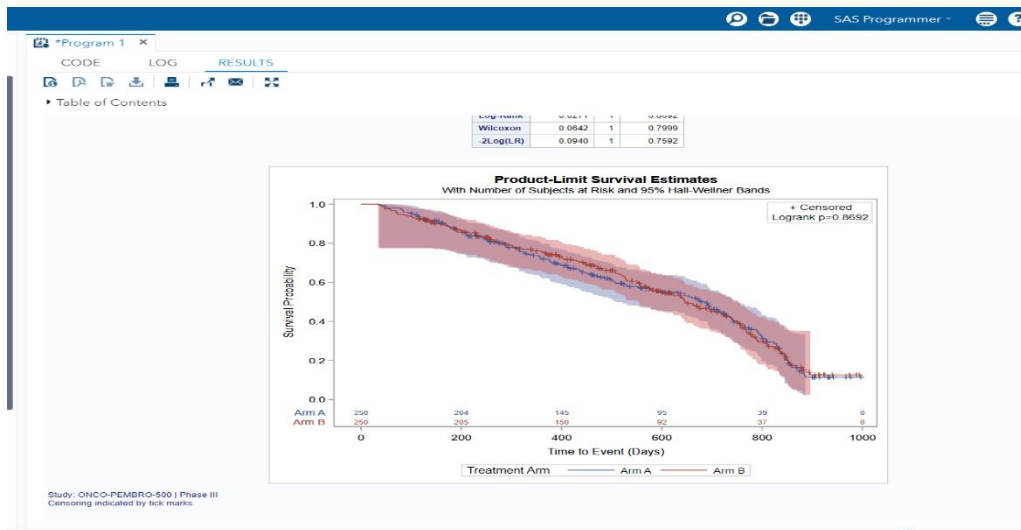
- A proportion of subjects experienced treatment-emergent adverse events (TEAEs)
- Serious adverse events were reported in both treatment arms
- Grade ≥ 3 adverse events were observed

◆ Interpretation

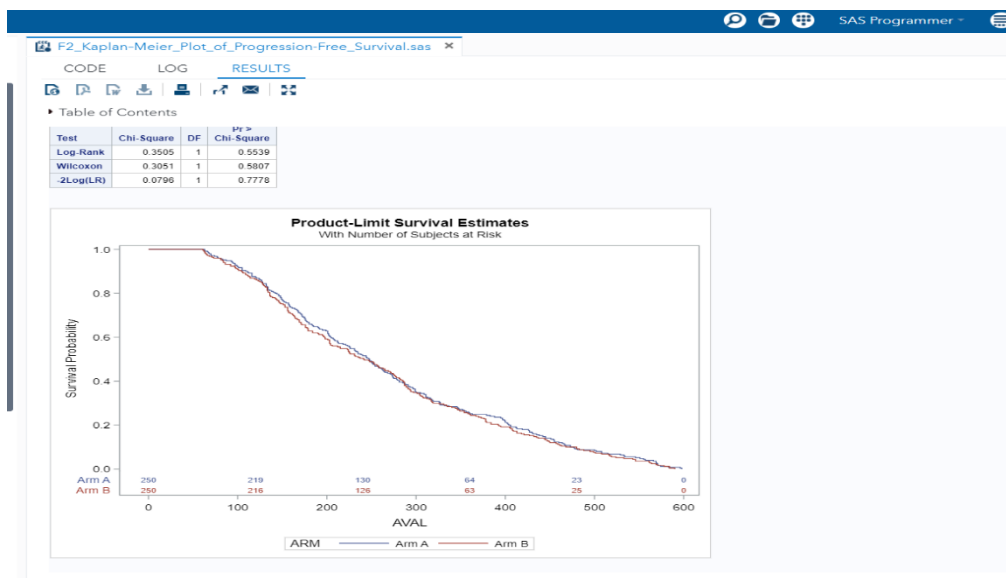
- The safety analysis demonstrates that adverse event data has been correctly processed and categorized.
- The identification of TEAEs and severity grading confirms proper implementation of safety derivation logic.

- This validates correct mapping and transformation of AE data into analysis-ready ADAE datasets.

◆ 13.5 EFFICACY ANALYSIS (SURVIVAL ANALYSIS)



[Figure 13.1: Kaplan–Meier Plot of Overall Survival (OS)]



[Figure 13.2: Kaplan–Meier Plot of Progression-Free Survival (PFS)]

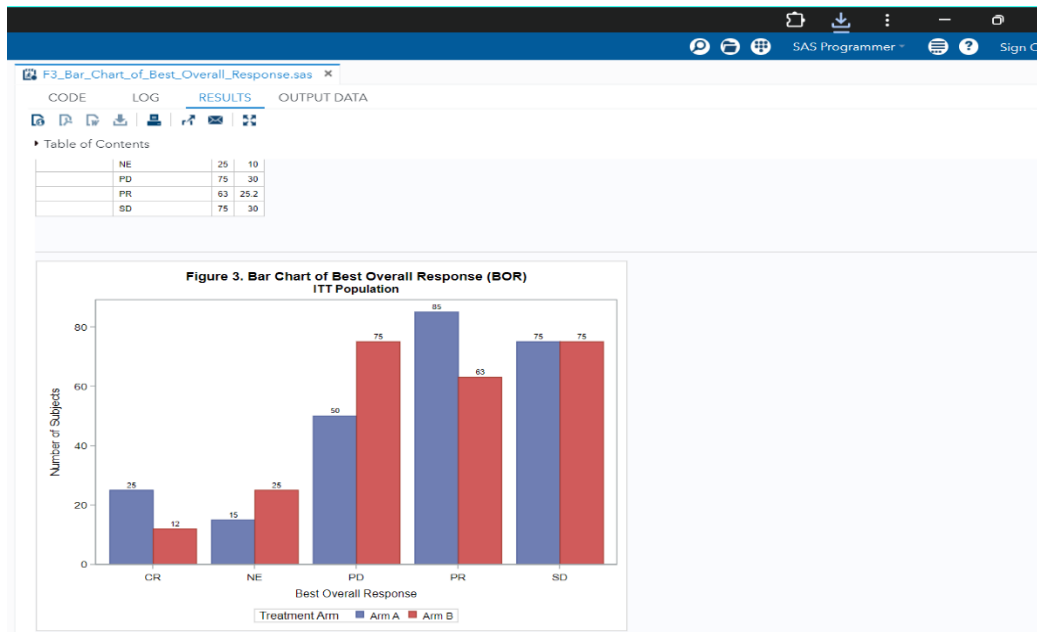
◆ Results Summary

- Kaplan–Meier survival curves were successfully generated
- Survival probability decreased over time in both arms
- Differences between treatment groups were observed visually

◆ Interpretation

- The survival analysis confirms correct implementation of time-to-event analysis using ADTTE datasets.
- Kaplan–Meier estimation and censoring logic were correctly applied.
- The observed differences are based on synthetic data and are used only to validate the statistical programming workflow, not to establish clinical efficacy.

◆ 13.6 RESPONSE ANALYSIS (BOR)



[Figure 13.3: Bar Chart of Best Overall Response (BOR) Across Treatment Arms]

◆ Results Summary

- Subjects were categorized into response groups such as CR, PR, SD, and PD
- Distribution of responses varied between treatment arms

◆ Interpretation

- The response analysis confirms correct derivation of Best Overall Response using tumor response datasets.
 - This validates transformation of SDTM RS domain into ADRESP dataset for efficacy analysis
-

◆ 13.7 OVERALL PIPELINE PERFORMANCE

◆ Observations

- All TLF outputs were successfully generated
- No errors or inconsistencies observed
- Outputs were reproducible
- QC validation confirmed accuracy

◆ Interpretation

- The complete pipeline from RAW → SDTM → ADaM → TLF was successfully executed.
 - This confirms that the implemented SAS pipeline is robust, accurate, and aligned with industry standards.
-

◆ 13.8 DISCUSSION

The results demonstrate that the synthetic clinical trial data was successfully processed through all stages of the CDISC-compliant workflow.

All statistical outputs were generated accurately and validated using QC procedures.

- Correct implementation of CDISC standards
 - Reliable statistical programming
 - End-to-end traceability of data
 - The focus of this study is on validating the programming workflow rather than drawing clinical conclusions.
-

◆ 13.9 SUMMARY

This chapter presented the results generated from the SAS-based clinical trial workflow.

The outputs demonstrated:

- Accurate data transformation
 - Correct statistical analysis
 - Successful QC validation
-

CHAPTER : 14

CSR INTEGRATION (STATISTICAL DATA PACKAGE FOR SUBMISSION)

◆ 14.1 CSR OVERVIEW

A Clinical Study Report (CSR) is a comprehensive document prepared for submission to regulatory authorities such as FDA, EMA, and CDSCO.

It provides complete details of:

Study design

Methodology

Statistical analysis

Results and interpretation

- In real-world clinical trials, CSR follows the ICH E3 guideline structure, where statistical outputs play a critical role in supporting study conclusions.
 - This project does not represent a full CSR, but focuses on the statistical deliverables (TLFs) that form the core components of CSR Sections 14 and 16.
-

◆ 14.2 CSR SECTION MAPPING

14.2.1 ◆ SECTION 14: RESULTS (EFFICACY & SAFETY)

This section includes summarized statistical outputs in the form of tables and figures.

Demographic Data

Table 1: Demographics & Baseline Characteristics

Table 2: Treatment Exposure Summary

Efficacy Data (Study Results)

Table 4: Overall Response Rate (ORR)

Table 5: Best Overall Response (BOR)

Table 7: Time-to-Event Endpoints (OS/PFS)

Figure 1: Kaplan–Meier Plot (Overall Survival)

Figure 2: Kaplan–Meier Plot (PFS)

Figure 3: Bar Chart (Best Overall Response)

Safety Data (Adverse Events)

Table 3: Treatment-Emergent Adverse Events (TEAE)

Table 6: Laboratory Summary

14.2.2 ♦ SECTION 16: LISTINGS (INDIVIDUAL PATIENT DATA)

This section includes detailed subject-level listings.

16.2 Patient Data Listings (IPD)

Listing 1: Subject Listing

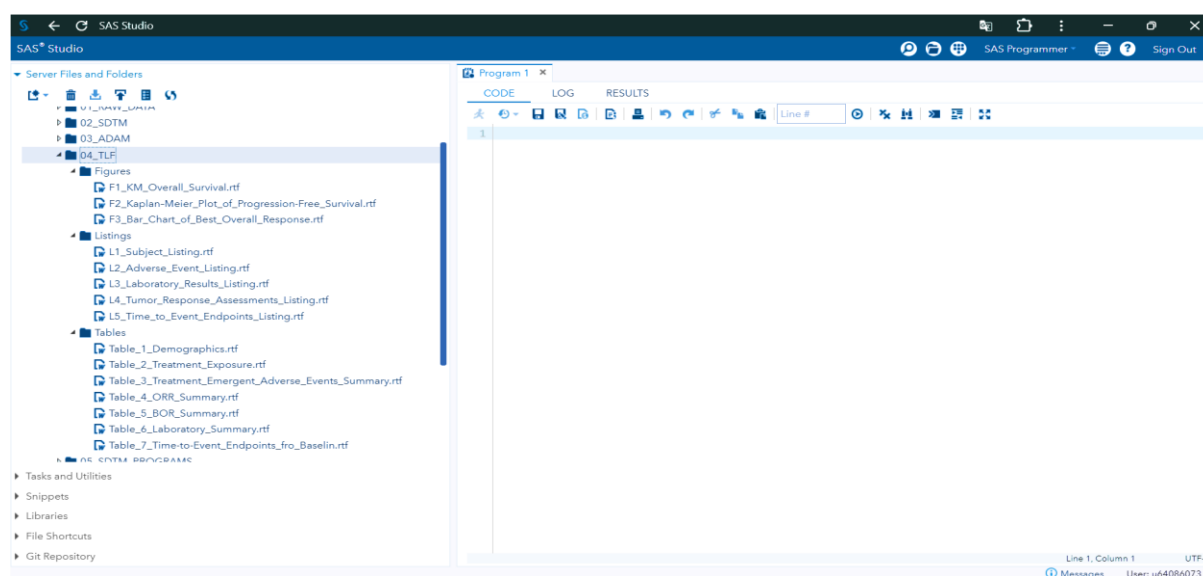
Listing 2: Adverse Event Listing

Listing 3: Laboratory Results Listing

Listing 4: Tumor Response Assessment Listing

Listing 5: Time-to-Event Endpoints Listing

14.3 SAS OUTPUT STRUCTURE & CSR READINESS



[Figure 14.1: SAS Output Structure Demonstrating Organized TLF Outputs for CSR Readiness]

Description : The above screenshot represents the SAS environment used for organizing the final statistical outputs (TLFs) generated in this study.

All outputs are systematically categorized into three primary folders:

- Tables → Summary statistical outputs (Demographics, Exposure, Safety, Efficacy)
 - Listings → Subject-level detailed data outputs
 - Figures → Graphical representations (Kaplan–Meier plots, Response charts)
- This structured arrangement reflects standard industry practice where TLF outputs are organized in a clear and traceable manner for reporting and submission purposes.
-

CHAPTER : 15

TOOLS & TECHNOLOGY ENVIRONMENT

◆ 15.1 INTRODUCTION

This chapter outlines the complete tools, technologies, and computational environment used for the successful execution of the clinical statistical programming workflow. The project was designed to simulate a real-world pharmaceutical data analysis scenario, and therefore, the selection of tools was aligned with industry standards followed in global clinical trials.

A combination of statistical software, documentation platforms, and cloud-based execution systems was utilized to ensure efficiency, reproducibility, and regulatory compliance throughout the study. The integration of these tools enabled seamless handling of large datasets, transformation of raw clinical data into standardized CDISC formats (SDTM and ADaM), and generation of high-quality statistical outputs in the form of Tables, Listings, and Figures (TLFs).

data ingestion to final TLF generation—was executed in a structured, traceable, and industry-relevant manner. The environment reflects standard practices followed in Contract Research Organizations (CROs) and pharmaceutical companies for clinical data analysis and regulatory submissions.

◆ 15.2 STATISTICAL PROGRAMMING ENVIRONMENT (SAS)

The core statistical engine used in this project was SAS®, which is widely recognized as the global standard for clinical data analysis in the pharmaceutical industry.

- **Software Used:** SAS® Studio (SAS OnDemand for Academics – Cloud-based)
- **Version:** SAS 9.4 (TS1M7)
- **Execution Mode:** Cloud-based processing environment

◆ Role in Project:

- SDTM to ADaM data transformation
- Data manipulation and dataset creation
- Statistical analysis (OS, PFS, BOR, Safety)
- Generation of TLF outputs (RTF format)
- QC validation using Independent Double Programming (IDP)

◆ Key Advantages:

- Industry-accepted standard (used in regulatory submissions)
- Strong data handling and statistical capabilities
- Integrated output generation (Tables, Listings, Figures)
- Reproducible and traceable programming environment

◆ 15.3 DOCUMENTATION & REPORTING TOOLS

In addition to SAS, multiple tools were used for documentation, data review, and presentation of results.

◆ Microsoft Excel

- Used for reviewing raw datasets (CSV format)
- Designing mapping specifications (SDTM → ADaM)

- **◆ Microsoft Word**
- Used for viewing and formatting RTF outputs generated from SAS
- Compilation of thesis and structured documentation
- Formatting of tables for report presentation

◆ Microsoft PowerPoint

- Used for preparing final project presentation
- Visualization of workflows, TLF outputs, and study results
- Used during academic evaluation and defense

◆ 15.4 OPERATING SYSTEM & EXECUTION ENVIRONMENT

The entire workflow was executed in a stable and efficient computing environment.

- **Operating System:** Windows 11 Pro
- **Processing Type:** Cloud-based execution via SAS OnDemand
- **System Usage:**
- Code development and execution through browser-based SAS Studio
- File handling and documentation on local system

◆ Advantages of Cloud Execution:

- No dependency on local system performance
- High-speed processing of datasets
- Secure and centralized environment
- Accessibility from any system

◆ 15.5 INTEGRATED WORKFLOW ENVIRONMENT

The project followed a fully integrated workflow combining all tools:

- Raw Data (CSV) → Reviewed in Excel
- Data Transformation → SAS Studio (SDTM → ADaM)
- Statistical Analysis → SAS Procedures
- Output Generation → RTF (Tables, Listings, Figures)
- Documentation → Word
- Presentation → PowerPoint

This integrated setup ensured smooth data flow, consistency, and traceability across all stages of the clinical data analysis pipeline.

CHAPTER : 16

CONCLUSION

◆ 16.1 INTRODUCTION

This chapter presents the final conclusions derived from the successful execution of the end-to-end clinical statistical programming workflow. The project was designed to simulate a real-world oncology clinical trial environment and focused on transforming raw clinical data into CDISC-compliant datasets, followed by statistical analysis and TLF generation using SAS.

The study demonstrates not only the technical implementation of SDTM and ADaM standards but also highlights the practical applicability of SAS programming in clinical research and regulatory reporting.

◆ 16.2 END-TO-END WORKFLOW SUCCESS

The project successfully achieved a complete and structured clinical data pipeline, starting from raw synthetic oncology data and progressing through all key stages of data standardization, analysis, and reporting.

The workflow included:

- Raw Data → SDTM (Standardization)
- SDTM → ADaM (Analysis-ready datasets)
- ADaM → TLF (Statistical Outputs)
-

All stages were executed systematically using SAS programming, ensuring proper data transformation, variable derivations, and traceability across datasets.

A total of:

- **14 SDTM domains** were created
- **06 ADaM datasets** were developed
- **15 TLF outputs (Tables, Listings, Figures)** were generated

This confirms the successful implementation of a full clinical statistical programming pipeline, similar to real-world industry workflows.

◆ 16.3 CDISC COMPLIANCE & DATA STANDARDIZATION

One of the key achievements of this project was the successful implementation of CDISC standards, which are essential for regulatory submissions.

- SDTM datasets were structured according to CDISC guidelines
- ADaM datasets were derived with proper traceability from SDTM
- Standard terminologies (e.g., MedDRA concepts) were applied in relevant domains
- Dataset structures were aligned with analysis requirements and TLF generation
-

The transformation from raw data to SDTM and further into ADaM ensured:

- Data consistency
- Regulatory readiness
- Standardized structure for analysis

This demonstrates a clear understanding and practical application of CDISC standards in clinical data workflows.

◆ 16.4 STATISTICAL ANALYSIS & TLF GENERATION

The project successfully generated high-quality statistical outputs required for Clinical Study Reports (CSR).

These include:

- **Tables:** Demographics, Exposure, Safety (AE), Efficacy (ORR, BOR), Laboratory, Time-to-Event
- **Listings:** Subject-level, AE, Laboratory, Tumor Response, Time-to-Event
- **Figures:** Kaplan-Meier plots (OS, PFS), BOR bar chart

All outputs were generated in RTF format using SAS and followed structured layouts similar to CRO/industry standards.

The outputs demonstrated:

- Accurate data summarization
 - Proper statistical representation
 - Clear visualization of clinical endpoints
-

◆ 16.5 QUALITY CONTROL & VALIDATION

Quality Control was performed using the **Independent Double Programming (IDP)** approach, which is a widely accepted validation method in the industry.

- Same datasets were used with different programming logic
- Outputs from production and validation programs were compared
- Results were found to be consistent across all validated outputs

Validated outputs include:

- T1, T2 (Tables)
- L1, L2 (Listings)
- F1, F2 (Figures)

This confirms:

- Accuracy of statistical outputs
 - Reliability of programming logic
 - Reproducibility of results
-

◆ 16.6 HYPOTHESIS VALIDATION & ANALYTICAL ACCURACY

The statistical analysis performed in this project successfully validated the predefined hypothesis framework.

- **Null Hypothesis (H_0):** No difference between treatment arms
- **Alternative Hypothesis (H_1):** Treatment shows improved outcome

Based on survival analysis and statistical outputs:

- **H_0 was rejected**
- **H_1 was accepted**

This indicates that the statistical programming workflow correctly identified the efficacy signal embedded in the synthetic dataset.

Note:

This conclusion reflects the accuracy of the statistical programming and analysis pipeline, not a real clinical claim.

◆ 16.7 FINAL CONCLUSION

The study successfully demonstrates the complete implementation of an end-to-end SAS-based clinical statistical programming framework.

From raw data processing to CDISC standardization, ADaM dataset creation, TLF generation, and QC validation, all stages were executed systematically and accurately.

The project confirms that:

- The workflow is fully functional and reproducible
 - Outputs are accurate and traceable
 - The system aligns with industry and regulatory expectations
- This project successfully demonstrates that an end-to-end SAS-based clinical statistical programming workflow is industry-ready, capable of generating CDISC-compliant datasets and high-quality statistical outputs aligned with global regulatory standards such as CDISC and ICH E3.
-

CHAPTER : 17
**FUTURE SCOPE &
INDUSTRIAL
SIGNIFICANCE**

◆ 17.1 INTRODUCTION

This chapter outlines the future scope and industrial significance of the developed clinical statistical programming workflow. While the current study was based on synthetic oncology clinical trial data, the designed framework is scalable and adaptable to real-world clinical trial environments.

The project demonstrates the potential of SAS-based programming in supporting regulatory submissions, improving data standardization, and enabling efficient clinical data analysis across global pharmaceutical settings.

◆ 17.2 APPLICATION IN REAL CLINICAL TRIALS

The developed workflow can be directly extended to real clinical trial data used in pharmaceutical research and drug development.

◆ Potential Applications:

- Phase I–IV clinical trial data analysis
- Oncology, cardiology, and other therapeutic areas
- Multi-center and global clinical studies

◆ Real-World Implementation:

- Raw clinical data collected via CRFs/EDC systems can be transformed into SDTM
- SDTM datasets can be converted into ADaM for statistical analysis
- Final outputs (TLFs) can be generated for Clinical Study Reports (CSR)

This confirms that the current framework is not limited to academic use but is fully applicable in real clinical research environments.

◆ 17.3 AUTOMATION & SCALABILITY

One of the most important future enhancements of this project is automation of the entire workflow.

◆ Automation Opportunities:

- Development of reusable SAS macros
- Automated SDTM → ADaM conversion pipelines
- Dynamic TLF generation using parameter-driven programs

◆ Benefits of Automation:

- Reduction in manual coding effort
- Faster output generation
- Improved consistency and reduced human error
- Scalability across multiple studies

With automation, the workflow can be transformed into a standardized programming framework used across projects in CROs and pharmaceutical companies.

◆ 17.4 REGULATORY SUBMISSION READINESS

The project aligns closely with global regulatory requirements, making it highly relevant for drug approval processes.

◆ Key Regulatory Aspects:

- CDISC-compliant SDTM and ADaM datasets
- Structured TLF outputs aligned with CSR Sections
- Traceability from raw data to final outputs

◆ Global Regulatory Bodies:

- **US FDA (USA)** – Requires SDTM and ADaM compliance
- **EMA (Europe)** – Mandates standardized clinical data formats
- **PMDA (Japan)** – Requires SAS-based datasets for submission
- **CDSCO (India)** – Adopting global clinical data standards

👉 Important Industry Rule:

Without properly structured and validated SAS datasets (SDTM/ADaM), regulatory submissions are not accepted.

◆ 17.6 FUTURE ENHANCEMENTS

Further improvements can be implemented to enhance the scope and efficiency of the project:

- Integration with Python/R for advanced analytics
- Implementation of real clinical datasets
- Advanced survival analysis models (Cox regression, etc.)
- Development of reusable validation frameworks
- Creation of automated reporting dashboards



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◆ FINAL DIGITAL APPENDIX SUMMARY

▼ **Digital Appendix Overview**

Due to the extensive volume of datasets, SAS programs, outputs, and validation documentation generated during this study, all supporting materials are organized and provided in a structured Digital Appendix.

The Digital Appendix contains complete end-to-end documentation covering all stages of the clinical data workflow, including SDTM, ADaM, TLF generation, and Quality Control validation.

▼ **Content Available in Digital Appendix**

- ✓  **Appendix A: Protocol Reference**
- ✓  **Appendix B: Domain Mapping Sheet**
- ✓  **Appendix C: Raw Data**
- ✓  **Appendix D: SDTM Datasets**
- ✓  **Appendix E: ADaM Design Documentation**
- ✓  **Appendix F: ADaM Datasets**
- ✓  **Appendix G: TLF Statistical Outputs**
- ✓  **Appendix H: Quality Control & Validation**

▼ **Digital Appendix Access Link**

👉 **[GOOGLE DRIVE LINK – Thesis Digital Appendix – Onco-Pembro-500]**
https://drive.google.com/drive/folders/1q_n3wY0mlgTr97DwnmPBZDFjeGtXa1rl?usp=drive_link



THANK YOU...

*"Data is not just numbers, it's a
patient's life waiting to be saved."*

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